

Master Thesis

Exploiting *k*-mers for genotype-phenotype association in spinach

submitted by

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in the framework of the Master programme

Biotechnology

in partial fulfilment of the requirements for the academic degree

Diplom-Ingenieur

Vienna, March 2025

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Affidavit

I hereby declare that I have authored this master thesis independently, and that I have not used any assistance other than that which is permitted. The work contained herein is my own except where explicitly stated otherwise. All ideas taken in wording or in basic content from unpublished sources or from published literature, as well as those which were generated using artificial intelligence tools, are duly identified and cited, and the precise references included.

I further declare that this master thesis has not been submitted, in whole or in part, in the same or a similar form, to any other educational institution as part of the requirements for an academic degree.

I hereby confirm that I am familiar with the standards of Scientific Integrity and with the guidelines of Good Scientific Practice, and that this work fully complies with these standards and guidelines.

Vienna, March 27, 2025

Lukas SCHÖNMANN (*manu propria*)

This is water.

—David Foster Wallace

Acknowledgements

I would like to express my deepest gratitude to my supervisors Assoc. Prof. Juliane Dohm and Prof. Heinz Himmelbauer for their invaluable feedback, consistent support on my scientific journey, and giving me the opportunity and freedom to pursue my research. I would also like to thank my colleagues at the Institute of Computational Biology, especially Felix, Thomas, Gil, Lukas, and Christian, who have been an inspiration to me and have provided insightful input.

I am immensely grateful to my family for their unconditional support, both emotionally and financially. Without your help, none of this would have been possible—words cannot express my gratitude. Special thanks also to my girlfriend for always encouraging and believing in me.

Lastly, I would like to acknowledge my friends for the meaningful discussions, laughter, and shared adventures. Your unique personalities and companionship have enriched my life in ways that go beyond academics.

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Abstract

Since its invention decades ago, DNA sequencing has become an essential tool in genomics. Modern high-throughput sequencing methods generate vast amounts of data, allowing large-scale sequencing of whole genomes and populations. Such data allows researchers to investigate genome-wide effects and identify important genes linked to specific traits. Genome-wide association studies (GWAS) are often used to detect genotype-phenotype associations based on single-nucleotide variation (SNVs) derived from read mapping and variant calling. In this study, both a traditional SNV-based GWAS and a modified k -mer-based GWAS were performed and compared, using Illumina whole-genome sequencing data from 303 cultivated and wild spinach (*Spinacia oleracea*) accessions. Spinach is a globally cultivated leafy vegetable with high economic importance, valued for its nutritional content. However, its production is threatened by downy mildew (DM), a disease caused by *Peronospora effusa*. The k -mer-based GWAS performed in this thesis successfully identified significant associations between genomic regions and key agronomic traits, notably detecting robust signals linked to DM resistance near chromosome 3. A gene involved in cuticular wax synthesis, located on chromosome 2 and previously unknown to be associated with DM resistance, emerged as a promising target for future breeding efforts. As part of this study, a specialized tool was developed which facilitates k -mer downstream processing. Random forest machine learning models were generated in this work and further proved the utility of k -mer data, predicting DM resistance phenotypes with an accuracy of 98 %. The k -mer-based approach demonstrated considerable potential to uncover genotype-phenotype associations. However, k -mer downstream analysis remains challenging due to the difficulties in linking k -mers to their genomic positions and a lack of optimized tools.

Kurzfassung

Seit ihrer Erfindung hat sich die DNA-Sequenzierung zu einem unverzichtbaren Instrument der Genomik entwickelt. Moderne Sequenzierungsmethoden erzeugen riesige Datenmengen und ermöglichen die Sequenzierung ganzer Genome und Populationen. Diese Daten können analysiert werden, um wichtige Gene zu identifizieren, die mit bestimmten Merkmalen verbunden sind. Dafür werden häufig genomweite Assoziationsstudien (GWAS) eingesetzt, die Einzelnukleotidsubstitutionen (SNVs) verwenden. In dieser Arbeit wurden eine traditionelle, auf kurzen Varianten basierende GWAS und eine neuartige, auf k -mers basierende GWAS anhand von Illumina-Ganzgenom-Sequenzierungsdaten von 303 kultivierten und wilden Spinat-Akzessionen (*Spinacia oleracea*) ausgeführt und verglichen. Spinat ist ein weltweit angebautes Blattgemüse mit hoher wirtschaftlicher Bedeutung, dessen Produktion durch Falschen Mehltau (*Peronospora effusa*) bedroht wird. Die k -mer-basierte GWAS identifizierte signifikante Assoziationen zwischen genomischen Regionen und wichtigen agronomischen Merkmalen, wobei starke Signale im Zusammenhang mit Mehltau-Resistenz in der Nähe von Chromosom 3 entdeckt wurden. Ein Gen auf Chromosom 2, das an der Cuticula-Wachssynthese beteiligt ist und bisher nicht mit DM-Resistenz in Verbindung gebracht wurde, stellt ein vielversprechendes Ziel für die Züchtung dar. Im Rahmen dieser Arbeit wurde ein spezielles Tool entwickelt, das die k -mer-Verarbeitung erleichtert. Mithilfe von k -mer-Daten wurden Random-Forest-Modelle für maschinelles Lernen erstellt, die Mehltau-Resistenz-Phänotypen mit einer Genauigkeit von 98 % voraussagen konnten. Der k -mer-basierte Ansatz zeigte ein beträchtliches Potenzial zur Aufdeckung von Genotyp-Phänotyp-Assoziationen. Die nachgelagerte k -mer-Analyse bleibt jedoch aufgrund der Schwierigkeiten bei der Verknüpfung von k -mers mit ihren genomischen Positionen und dem Mangel an optimierten Tools eine Herausforderung.

1 Introduction

1.1 Genomic sequencing

Since the discovery of deoxyribonucleic acid (DNA) and its purpose of storing the information necessary for cell development, functioning, and reproduction, there was an aspiration to obtain accurate sequences of complete genomes. Among the first sequencing methods was Sanger sequencing, published in 1977 (Sanger et al., 1977). It became the gold standard for many years, producing accurate sequencing reads of up to 1000 base pairs (bp) length. Although its automation enabled decoding the first complete human genome sequence (International Human Genome Sequencing Consortium, 2004), it was limited by throughput and cost (Schadt et al., 2010).

About 30 years later, these first-generation sequencing methods became superseded by next-generation sequencing (NGS) methods leading to a spectacular yearly decrease in DNA sequencing costs—much bigger than predicted by scaling laws (Wetterstrand, 2025).

A commonly used NGS method is sequencing by synthesis: first, the sample DNA is fragmented into short pieces. Then, special adapters are ligated to both ends of the template fragments, and the fragments are attached to a surface where they are amplified. Typically, fluorescently labeled nucleotides are added enzymatically one at a time. The fluorescent signal is detected and translated into a DNA sequence (McCombie et al., 2019). This process can be massively parallelized by sequencing millions to billions of clusters simultaneously, producing a huge amount of data at low costs. Popular platforms utilizing this method are provided by Illumina, providing billions of short reads (up to 150 bp paired-end reads) per run with high accuracy (K. R. Kumar et al., 2019).

While NGS technologies excel at producing accurate short reads, third-generation sequencing methods like single-molecule real-time (SMRT) sequencing from Pacific Biosciences (PacBio) introduced long-read sequencing. SMRT sequencing can be used to generate reads above 1,000 bp length, reaching up to 10,000 bp. Additionally, it can capture methylation patterns along with the nucleotide sequence. Although SMRT sequencing has a higher error rate than NGS methods, advanced consensus algorithms have improved its accuracy (Schadt et al., 2010).

While the high accuracy and throughput makes short reads ideal for applications requiring precise variant calling, such as genome-wide association studies (GWAS), long reads are particularly useful at resolving complex genomic regions, such as repeats or structural variants.

Hybrid sequencing approaches utilize the advantages of both methods for high-quality de novo genome assembly (Rhoads & Au, 2015).

1.2 *K*-mers in genomics

The vast amounts of sequencing data generated by both short- and long-read sequencing technologies require efficient computational strategies. Many applications in bioinformatics involve the use of *k*-mers—short substrings of length *k* from a DNA or RNA sequence. For example, a sequence can be broken down in contiguous 31-letter subsequences (31-mers) starting from each position of the sequence (Figure 1). A sequence of length *L* contains up to $L - k + 1$ *k*-mers and n^k total possible *k*-mers, where *n* is the alphabet size. Since it is unknown which strand of the double-stranded DNA has been sequenced, each DNA *k*-mer is usually represented in a normalized orientation: the lexicographically smaller between the *k*-mer and its reverse complement is chosen as the canonical form (Figure 2). In many applications, only the canonized *k*-mers are used, ensuring that double-stranded DNA is accounted for without redundancy (Marçais et al., 2024). Frequently, odd values are selected for *k*, as the character in the middle of the string helps resolve ambiguities when encountering sequences where the original sequence and its reverse complement are identical.

Sequence	A A C T A G T C T T A G A A T A C T A C C A A A C A T T G T G G T G G G T
1 st <i>k</i> -mer	A A C T A G T C T T A G A A T A C T A C C A A A C A T T G T G
2 nd <i>k</i> -mer	A C T A G T C T T A G A A T A C T A C C A A A C A T T G T G G
3 rd <i>k</i> -mer	C T A G T C T T A G A A T A C T A C C A A A C A T T G T G G T
4 th <i>k</i> -mer	T A G T C T T A G A A T A C T A C C A A A C A T T G T G G T G
5 th <i>k</i> -mer	A G T C T T A G A A T A C T A C C A A A C A T T G T G G T G G
6 th <i>k</i> -mer	G T C T T A G A A T A C T A C C A A A C A T T G T G G T G G G
7 th <i>k</i> -mer	T C T T A G A A T A C T A C C A A A C A T T G T G G T G G G T

Figure 1: Illustration of overlapping *k*-mers of length 31 generated from a nucleotide sequence. The 37 bases long sequence contains $37 - 31 + 1 = 7$ *k*-mers.

In recent years, methods that utilize *k*-mers, have become more important in the field of bioinformatics (Moeckel et al., 2024). *K*-mer-based approaches have found a wide range of applications in medicine, agriculture, or basic research. The frequencies of *k*-mers in sequences provide valuable information on structural variations (Sohn et al., 2023) or repetitive elements (Titievsky et al., 2021). The characteristics of *k*-mer spectra can reveal taxonomic insights (Chor et al., 2009) and can be used to identify and remove sequencing read errors (Liu et al., 2013). *K*-

mer-based algorithms are employed in genome assembly and sequence alignment (H. Li, 2018). In comparative genomics, k -mers leverage the construction of phylogenetic trees (Ondov et al., 2016). Additionally, extending k -mers to protein sequences has shown success in predicting folding patterns (Santoni, 2021). Some methods can be enhanced by substituting conventional approaches with k -mers. For instance, it has been suggested that k -mer-based genome-wide association studies (GWAS) can overcome limitations of traditional designs by selecting k -mers instead of single-nucleotide polymorphisms as molecular markers to find associations between complex traits and genetic variants (Lemay et al., 2023; Voichek & Weigel, 2020).

Sequence	A	C	G	T	A	A	T	G	C
Complement	T	G	C	A	T	T	A	C	G
Reverse complement	G	C	A	T	T	A	C	G	T
Canonical k -mer	A	C	G	T	A	A	T	G	C

Figure 2: Demonstration of canonical k -mer determination. Shown are a DNA sequence, its complementary strand, and the reverse complement. The canonical form of the k -mer is the lexicographically smaller between the sequence and its reverse complement.

1.3 Genome-wide association studies in plants

A genome-wide association study (GWAS) is an approach to scan the entire genome for genetic variants (often single-nucleotide polymorphisms, SNPs) that are statistically associated with a certain trait or disease. In a GWAS, thousands or millions of genetic variants spread across the genome are genotyped in a population of individuals, and each variant is tested for correlation with the phenotype of interest.

GWAS analyses typically use statistical models, such as linear regression, to test each genetic variant for association while correcting for multiple testing. One key challenge in early GWAS was controlling for population structure and relatedness between samples, which can confound associations for variants that are more common in a subpopulation when a particular phenotype is more pronounced in that group. To address this, linear mixed models (LMMs) have been adopted as a standard method in GWAS. LMM-based methods include a fixed effect that captures population structure and a random effect of kinship, thereby accounting for population stratification within the data (Tibbs Cortes et al., 2021).

Since their advent more than 15 years ago, genome-wide association studies have seen great success in identifying genetic variants associated with complex traits and diseases. They have not only been effectively applied to clinical research, epidemiology, and drug testing (Abdellaoui et al., 2023) but also found success in crop improvement. Marker-trait associations identified through GWAS have been utilized for marker-assisted selection in crop plants, including rice,

soybean, wheat, and maize. Among the traits studied are biotic and abiotic stress resistances, nutritional and morphological traits, and quality parameters (Gupta et al., 2019). A landmark example of GWAS results has been the identification of rare alleles in maize (*Zea mays*), which enabled biofortification with pro-vitamin A and helped reduce the prevalence of vitamin A deficiency and related complications (Xiao et al., 2017).

However, the results of genetic association studies have not been without limitations. Controversies revolve mainly around spurious associations due to population structure or multiple causative factors, correlations between traits and the environment (Platt et al., 2010), difficulties in tracing the signals to the actually underlying causative variants (Dickson et al., 2010), explaining only a small fraction of total genetic contribution to complex traits, leaving “missing” heritability unexplained (Manolio et al., 2009), and the challenge of translating GWAS results into biological effects—“the dark road from association to function” (Edwards et al., 2013). Traditional GWAS also primarily focus on individual genotype-phenotype associations under the assumption of an additive model and do not inherently account for epistasis, i.e., interactions between variants (Guindo-Martínez et al., 2021).

Nevertheless, over time, GWAS have established themselves as a cornerstone of genetic research, not only in human medicine but also in crop plant breeding (Tibbs Cortes et al., 2021).

1.4 Machine learning and random forests

Machine learning (ML) is the field of artificial intelligence focused on developing algorithms that can learn from data through pattern recognition, automatically enhancing their performance by repeatedly fitting a model and making corrections to improve it. Rather than being explicitly programmed for a specific task, ML models are capable of adapting to patterns in the training data and use them to make predictions or decisions. This makes them especially suited for analyzing biological datasets, which are often both large and complex (Greener et al., 2022).

Broadly, ML methods can be divided into supervised learning (where data has been labelled, e.g. with experimental measurements) and unsupervised learning (identifying hidden structure in unlabeled data). A loss function measures the discrepancy between the output and ground truth; model optimization is achieved by minimizing it. Hyperparameters are a set of values that must be defined when instantiating a model, stay constant during training, and shape how the adjustable parameters of the model are modified during training. Finding the optimal hyperparameters is challenging but greatly impacts model performance (Greener et al., 2022).

A common pitfall in machine learning is introducing bias by overfitting the model to the data. An overfitted model entirely memorizes the data it was trained on, achieving great results on the known data but failing when being applied to new information. For this reason, it is essential to separate the input data into a train and test set. The model must be strictly trained on the training set, while the final model performance is evaluated on the test set only. Often, the training set is further split into subsets using cross-validation, rotating over the partitions and using one as a validation set for hyperparameter optimization (Greener et al., 2022).

Machine learning has been extensively applied to several fields in biology to solve complex problems on large-scale data. In genomics and proteomics, ML methods see use in function and structure prediction, motif identification, and linkage analysis. Further topics include microarray analysis, pathway and network modelling in systems biology, phylogenetic tree construction, image analysis, and text mining. Some of the techniques involved in machine learning are Bayesian classifiers, discriminant analysis, neural networks, hidden Markov models and Gaussian networks to perform tasks such as classification, regression, or clustering (Larrañaga et al., 2006).

A powerful ML technique used across many scientific domains is the random forest (RF) algorithm. Random forests are an ensemble learning method that builds a multitude of decision trees and aggregates their results for more accurate and stable predictions. In a random forest, each individual decision tree is trained on a random subset of the data and, for each split, a random subset of the features. The average of majority vote is used to estimate a more accurate prediction (Breiman, 2001). By combining the results of many trees, random forests tend to reduce the problem of overfitting and generalize well. Their high accuracies, robustness to noise, and ability to handle a large number of features make them well-suited for biological data (Breiman, 2001; X. Chen & Ishwaran, 2012; Salman et al., 2024), and they have been utilized for genetic association studies where they can capture complex interactions among variants (Goldstein et al., 2011).

1.5 Spinach: genetics, cultivation, and economic importance

Spinach (*Spinacia oleracea*) is an economically important leafy green crop native to Asia but now cultivated worldwide. It is a diploid ($2n = 2x = 12$), highly heterozygous plant belonging to the *Amaranthaceae* family in the Caryophyllales order. Spinach is mostly dioecious, meaning that there are separate male and female plants, and is pollinated by wind. Male plants have two different kinds of sex chromosomes (XY) while females are homogametic (XX). Recent research suggests that chromosomes 4 of male and female genomes are the sex chromosomes (She et al.,

2020). Specific environmental conditions can induce sex reversion, enabling selfing in spinach. It is an annual plant, with plants moving from the vegetative to the reproductive stage in late spring or summer (Bhattarai & Shi, 2021).

Wild spinach accessions (*S. tetrandra* and *S. turkestanica*) serve as a valuable genetic reservoir and have been used extensively for introgressing desirable quality traits and disease resistances in crop breeding (Bhattarai & Shi, 2021). Due to its nutritional content and health benefits, spinach production has risen steadily since 1980 (Food and Agriculture Organization [FAO], 2024) and this trend is expected to continue in the coming years. Spinach thrives in cool seasons and has a global production of around 30 million tons, with China alone accounting for over 90 % of the total output (Bhattarai & Shi, 2021).

A wealth of data has been published on spinach, with a chromosome-scale, high-quality reference genome of the Monoe-Viroflay cultivar made available by Cai et al. (2021), along with a genome annotation and re-sequencing data of 305 cultivated and wild spinach accessions. Monoe-Viroflay is a highly homozygous inbred line, developed by repeatedly selfing a single monoecious plant (containing both male and female parts).

1.6 Downy mildew in spinach

Downy mildew (DM) represents the most serious disease threat on spinach production, drastically decreasing quality and quantity of spinach yields worldwide (J. C. Correll et al., 2011). This disease is caused by the obligate oomycete *Peronospora effusa*, a fungus-like eukaryotic microorganism. The pathogen infects spinach leaves and produces abundant wind-dispersed spores that can travel long distances and infest seed lots (Kandel et al., 2019). Infected plants develop yellow spots on the foliage, making the fresh product unmarketable (Klosterman, 2016).

The disease cycle of *P. effusa* includes both asexual reproduction (via sporangia that spread epidemically in the field) and sexual reproduction (via oospores, serving as primary inoculums). Through sexual recombination, the pathogen creates genetic diversity and may acquire new virulence (Kandel et al., 2019). As a result, the pathogen population is highly variable: to date, at least 20 distinct races of *P. effusa* have been identified (International Seed Federation [ISF], 2024). New pathogen strains continue to emerge, spurred on by using spinach lines relying on the same resistance mechanisms and increased organic production. These new races are often capable of overcoming the genetic resistance present in existing spinach cultivars (Bhattarai et al., 2022).

Managing downy mildew resistance in spinach relies heavily on breeding and deploying resistant varieties, especially in organic production. Typically, cultivars carry single-gene resistances (James C. Correll et al., 1994), and several resistance loci have been identified, such as the *RPF* loci on chromosome 3 (Bhattarai et al., 2022). However, this is problematic insofar as the pathogen has shown a remarkable ability to adapt. For instance, resistance genes that protected against earlier races of *P. effusa* may become ineffective as newly emerging races exhibit novel virulence patterns (J. C. Correll et al., 2011; ISF, 2024). This “arms race” between spinach and *P. effusa* makes breeding for durable broad-spectrum resistance a challenge.

Resistance breeding against newly emerging pathogen races remains difficult due to the limited understanding of resistance mechanisms and restricted genetic diversity among current cultivars. Nevertheless, given the substantial economic impact of the disease, developing sustainable genetic resistance strategies will stay highly relevant in the upcoming years.

1.7 Objectives of this study

This study aims to identify associations between genomic regions and key agronomic traits in spinach by conducting genome-wide association analyses, with a specific focus on downy mildew resistance. To achieve this, the traditional GWAS approach based on single-nucleotide variation (SNVs) derived from read mapping and variant calling is performed and compared to a *k*-mer-based association analysis, which is run and evaluated for its potential to overcome some of the limitations of the conventional GWAS approach. To assess the effectiveness of *k*-mer-based methods, this study seeks to replicate findings from previous GWAS using the same genomic raw data as input. Additionally, it addresses challenges in the downstream analysis of significantly associated *k*-mers, for which no satisfactory solutions have yet been developed, and explores new strategies to improve their interpretation. Furthermore, the study investigates whether *k*-mer data can effectively serve as input for random forest machine learning models to accurately predict resistance phenotypes, with the additional goal of identifying molecular markers relevant for spinach breeding.

2 Materials and methods

In this study, re-sequencing data from 303 spinach accessions, including cultivated (*S. oleracea*) and wild species (*S. tetrandra* and *S. turkestanica*), was utilized. These datasets were originally generated by Cai et al. (2021) using the Illumina HiSeq 4000 platform in paired-end mode, producing datasets with 2×150 base pair raw sequences in FASTQ format. Additionally, the authors constructed a high-quality reference genome based on the inbred line Monoe-Viroflay, assembled using a combination of PacBio continuous long reads and Illumina short reads. The study also provided phenotypic data for 20 important agronomic traits related to plant morphology, bolting & flowering behavior, disease resistance, and nutritional quality.

The raw sequencing reads were retrieved from the Sequencing Read Archive (SRA, Kodama et al., 2012), stored under the BioProject identifier PRJNA976336. Read archives were downloaded and extracted to FASTQ format using the `prefetch` and `fasterq-dump` tools from the SRA Toolkit (NCBI SRA Toolkit Development Team, 2024). The reference genome and gene annotation data were obtained from SpinachBase (Collins et al., 2019; *SpinachBase*, 2025). Detailed metadata on spinach accessions and phenotypic information can be found in Supplementary Data 6 of Cai et al. (2021).

2.1 *K*-mer-based pipeline

K-mer counting, filtering, generation of a table containing the *k*-mer presence/absence patterns, and the *k*-mer-based GWAS steps were all done using the `kmersGWAS` pipeline by Voichkek and Weigel (2020), version 0.3-beta.

2.1.1 *K*-mer counting and pre-processing

A *k*-mer length of 31 was chosen for all subsequent analyses. *K*-mer counts were obtained from the raw sequencing reads of the spinach accessions using `KMC` version 3.0.0 (Kokot et al., 2017). *K*-mers were counted twice: first, the canonized forms were counted (i.e., the lexicographically lower one of a *k*-mer sequence and its reverse complement), and a canonized *k*-mer had to appear at least twice in an accession to be counted. Then, *k*-mers were counted again without canonization, taking the *k*-mer as it was present in the raw sequencing data. Non-canon *k*-mer counts were not filtered. For each accession, the *k*-mer counts were then combined to a single list using the `kmers_add_strand_information` command from the `kmersGWAS` pipeline. Log

files of the k -mer counts created by KMC were analyzed and the data plotted using the Jupyter Notebook environment (Kluyver et al., 2016) and Python 3 (van Rossum & Drake, 2009).

For k -mer count saturation analysis, the raw sequencing reads of four accessions were downsampled from around 480-580 million reads to 100 million, 80 million, and 40 million reads using Seqtk version 1.0-r82-dirty, setting the seed to “100” via the `-s100` flag (H. Li, 2014).

The individual k -mer lists were then merged and filtered to form a single k -mer table, containing the presence/absence counts from all k -mers found in the raw sequencing data for all 303 accessions. For a k -mer to be included in the final table, it had to be present in at least five samples, corresponding to a minor allele count (MAC) of 5. Additionally, a k -mer had to appear in both its canonized and non-canonized form in at least 20 % of samples. Combining and filtering the lists was done using the `list_kmers_found_in_multiple_samples` command from the kmersGWAS pipeline with the parameters `--mac 5` and `-p 0.2`. To generate the table with the presence/absence patterns, the pipeline’s command `build_kmers_table` was executed.

A kinship matrix of the pairwise relatedness between accessions was calculated using the `emma_kinship_kmers` command with the parameter `--maf 0.015` from the kmersGWAS pipeline. Relatedness was calculated as in EMMA (Kang et al., 2008) but adapted to use k -mers instead of single-nucleotide polymorphisms (SNPs), so that each position in the matrix represents the proportion of shared k -mers a pair of accessions.

2.1.2 Genome-wide association testing using k -mers

To run a k -mer-based genome-wide association analyses for all 20 traits, the `kmers_gwas.py` Python script from the kmersGWAS pipeline was executed with the parameter `--kmers_number 100000`. This association analysis consisted of two steps: first, a fast but approximate association testing which reduced the number of k -mers in the dataset to 100,000. Then, exact association tests returning significantly associated k -mers were performed, with the significance threshold obtained from running the test on 100 permutations. Because of the high computational demands of the exact association tests on the large number of permuted phenotypes, it was necessary to reduce the size of the dataset via the initial pre-filtering step.

The pre-filtering step of the k -mer dataset calculated approximate scores for each k -mer by doing association testing based on the GRAMMAR-Gamma method (Svishcheva et al., 2012). The running time of this method scales linearly with the sample size and performed well even with the hundreds of millions of k -mers present in the input data. Only the top-scoring 100,000 k -mers were then retained for exact association testing in the second step. This step was also

performed by retaining 1,000 and 1,000,000 pre-filtered k -mers, yielding similar results. However, keeping only 1,000 top-scoring k -mers risked losing actually significant k -mers because it was close to the number of significant k -mers captured by the exact tests. On the other hand, 1,000,000 pre-filtered k -mers increased the computational burden by a large extent.

The exact association tests were performed with **GEMMA** (X. Zhou & Stephens, 2012). Because of an issue regarding the **kmersGWAS** pipeline, version 0.96 in the original pipeline had to be replaced with the updated version 0.98.3. To obtain significance level thresholds and control for the family-wise error rate (FWER), in the pipeline, the phenotypes were randomly permuted 100 times utilizing the **MVNpermute** algorithm implemented in R (Abney, 2015), and genome-wide association (GWA) tests were then conducted on each permutation. **GEMMA** linear mixed models (LMM) with the likelihood ratio test were used to calculate a p -value for each k -mer. The lowest p -value of each permutation was stored in a list. The $n\%$ family-wise error rate threshold was then obtained by taking the n th lowest p -value from this list of 100 p -values. This way, the 5 % and 10 % FWER significance level thresholds were calculated by taking the fifth and tenth lowest entries in the list of sorted p -values. Then, another genome-wide association test was performed on the unpermuted data, and significant k -mers were extracted which had p -values below the 5 % or 10 % FWER thresholds, respectively. For all subsequent analyses, only the k -mers with p -values below the 5 % FWER thresholds were used.

2.1.3 Downstream analyses of significant k -mers

For each trait, the raw sequencing read pairs containing any of the significant k -mers or their reverse complements were extracted. To facilitate efficient and flexible extraction of records from multiple terabytes of raw sequencing data based on matching k -mers, a specialized tool was developed. The program was written in the Rust programming language, version 1.82.0 (Rust Foundation, 2024), and allowed for quick k -mer processing while collecting detailed match statistics.

The extracted paired-end reads were then mapped to the Monoe Viroflay spinach reference genome using **Bowtie 2** version 2.5.1 with the `--very-sensitive-local` preset, which corresponds to the parameters `-D 20 -R 3 -N 0 -L 20 -i S,1,0.50 --local` (Langmead & Salzberg, 2012). In “local” mode, **Bowtie 2** allows soft clipping characters when aligning them, so that it is not required for an entire read to align from one end to the other. The aligned sequences were then sorted, and an index and mapping statistics were generated with **SAMtools** version 1.21 (Danecek et al., 2021). To identify aligned sequences that mapped to genic regions

annotated in the genome, **BEDtools** version 2.27.1 was used (Quinlan & Hall, 2010). Sequence mappings to the reference, along with the genome annotation, were then visualized in the **Integrative Genomics Viewer (IGV)** version 2.12.3 (Thorvaldsdóttir et al., 2013). Using the specialized tool, aligned reads were tagged with the presence of significant *k*-mers in concordance with the Sequence Alignment/Map Format Specification and Optional Fields Specification (The SAM/BAM Format Specification Working Group, 2024a, 2024b). The reads were then highlighted in IGV according to their tags.

Presence/absence patterns of significant *k*-mers were extracted from the *k*-mer table executing the `filter_kmers` command from the **kmersGWAS** pipeline. Then, heatmaps of the hierarchically clustered presence/absence patterns were generated using Python. Detailed match statistics were logged with the specialized tool during extraction of raw sequencing reads containing significant *k*-mers. These statistics were utilized to plot the lowest *p*-values of significant *k*-mers contained in extracted sequencing reads against their corresponding genomic mapping positions. These charts resembled a type of plot commonly used in SNP-based GWAS (Gibson, 2010) and were called *k*-mer-based Manhattan plots.

2.2 GWAS workflow based on read-mapping and variant calling

Variants called from the spinach raw sequencing data in the Variant Call Format (VCF) were sourced from a previous study authored by members of the Institute of Computational Biology (Nguyen-Hoang et al., 2024). They called single-nucleotide substitutions and small indels using **GATK** (Poplin et al., 2017), and applied the hard filtering approach, which included the following filtering criteria: $\text{MLEAF} > 0.01$, $\text{MQ} > 30.0$, $\text{DP} > 610$, $\text{DP} < 3050$, $\text{SOR} < 3.0$, $\text{QD} > 2.0$, $\text{FS} < 60.0$, $\text{MQRankSum} > -12.5$, $\text{ReadPosRankSum} > -8.0$, $\text{BaseQRankSum} > -12.5$. Only biallelic alleles were retained and the maximum allowed fraction of no-calls was set to 0.1.

2.2.1 Quality control and data cleaning

Summary statistics of the called variants were calculated using **BCFtools** version 1.20 (Danecek et al., 2021). Additionally, a quality control (QC) report was generated with the **VariantQC** tool, version 1.3.71 (Yan et al., 2019). Chromosome names in the VCF file had to be renamed and unscaffolded contigs removed to enable using this tool on the data.

Further data cleaning and QC steps were undertaken with the **PLINK** toolset, version 1.90b6.24 (Chang et al., 2015). Statistics for SNP and individual missingness, minor allele frequency (MAF), Hardy-Weinberg equilibrium (HWE), and heterozygosity were analyzed and the data

filtered according to commonly suggested values (Marees et al., 2018). Maximum SNP and individual missingness were set to 0.05. Variants with a HWE p -value below 10^{-6} were removed. Due to the large sample size, the MAF threshold was set to the lower value of 0.01. Only a few variants had extreme heterozygosity rates.

2.2.2 Genome-wide association testing using variant calls

PLINK was used to generate all necessary binary files for the GWAS, storing the genotype and sample information. Phenotype data was converted to the correct format required by PLINK. GEMMA version 0.98.5 was then used to calculate a mean-centered distance matrix, estimating the genetic relationship between pairs of accessions (X. Zhou & Stephens, 2012). To account for population stratification, i.e., the presence of subgroups with heterogeneous ancestral backgrounds, the centered relatedness matrix was used in the LMM calculations.

Association tests for all 20 agronomic traits were also performed with GEMMA linear mixed models, using the estimated relatedness matrix to correct for population structure. For each SNV, a p -value was calculated using the Wald test. To control the family-wise error rates, the unadjusted alpha level of 0.05 was corrected using the Bonferroni method: a genome-wide significance threshold is obtained by dividing α by the number of variants tested (Haynes, 2013). The Bonferroni-corrected p -value threshold was $0.05/3,610,120 = 1.38 \cdot 10^{-8}$.

Significant variants with p -values below this threshold were extracted to separate VCF files using BCFtools. Manhattan plots were generated using the QMplot Python module (Huang & Libiseller-Egger, 2022). The p -values of each variant were plotted against their corresponding position in the genome. Additionally, the genome-wide and suggestive significance threshold lines were drawn at the Bonferroni-adjusted values for alpha levels of 0.05 and 1, respectively.

The extracted significant variants were then functionally annotated using SnpEff version 5.1 (Cingolani et al., 2012). Because there was no pre-built database available for Spinach, a new one was constructed from the spinach reference genome. Subsequently, the variants were annotated with their predicted functional effects and effect size.

2.3 Machine Learning and dimensionality reduction

All machine learning models were trained on the presence/absence patterns of the pre-filtered set of 100,000 k -mers for the downy mildew resistance trait, obtained by the approximate association testing step of the k -mer-based GWAS. For the models, random forest classifiers were chosen, using the implementation from the Python module Scikit-learn (Fabian Pedregosa

et al., 2011). Data pre-processing, parameter tuning, model training and performance assessment were also performed with the Scikit-learn framework in a Jupyter Notebook environment. To avoid model overfitting, the data was split into train and test subsets, comprising 75 % (220 accessions) and 25 % (74 accessions) of the data, respectively. To account for the imbalanced distribution of the DM phenotype, stratified sampling was employed.

The model hyperparameters were tuned using grid search in conjunction with cross-validation. The grid search generated candidate models from all combinations of a range of parameters. Cross-validation was used to avoid bias and assess candidate performance. For each candidate, the training data was partitioned into five subsets (folds), using stratified sampling to retain the original phenotype distribution in the subsets. Candidate models were then trained five times, rotating over four training folds and one validation fold (Figure 3). The average of the model accuracies was used to determine the best hyperparameters from the parameter grid.

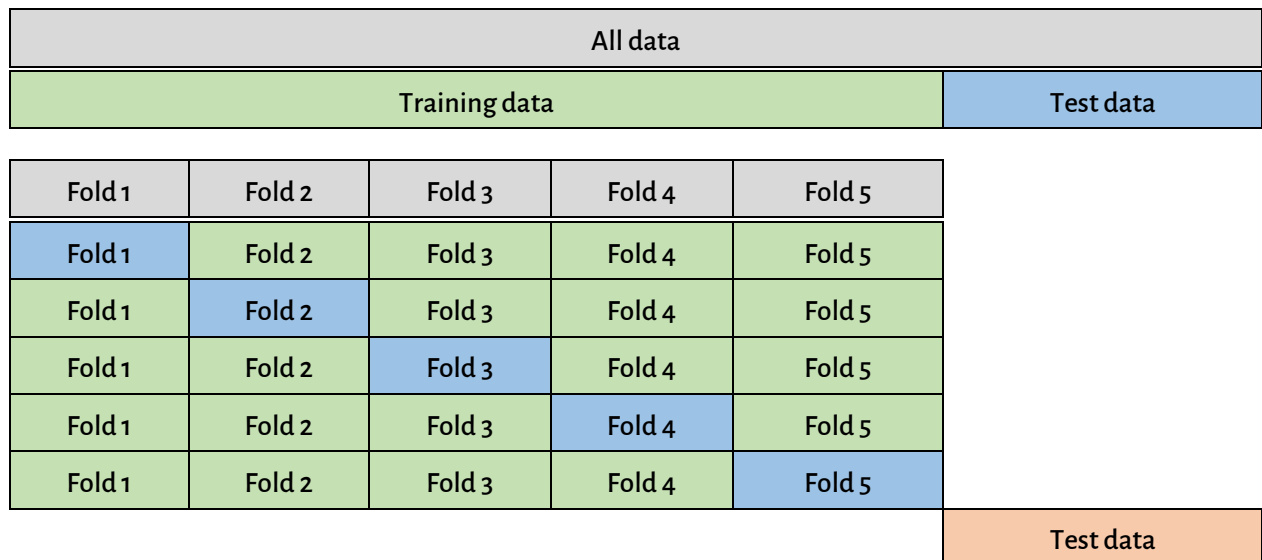


Figure 3: Schematic representation of the data splitting and cross-validation procedure. The dataset (“All data”) is first split into training data (green) and test data (blue). The training portion is then partitioned into k folds (Fold 1 through Fold 5), where each fold is used in turn as a validation set (blue cells) while the remaining folds (green cells) serve as the training set. After performance is averaged across all test folds to fine-tune model parameters, the final model is retrained on the entire training set and evaluated on the held-out test data (orange) for an unbiased performance estimate. Modified from the Scikit-learn User Guide (Scikit-learn developers, 2025).

Three different data transformations of the downy mildew resistance phenotype were used to obtain random forest models: “original”, “binary-split”, and “binary-polarized”. One RF model was trained on the original data as provided by Cai et al. (2021), except for dropping the single accession in the “resistance” class. In the “binary-split” approach, the class labels were recoded into a binary trait, grouping the labels “highly susceptible”—“susceptible” and “medium

resistance”—“resistance”—“high resistance” together to form a susceptible and resistant class, respectively.

The third strategy “binary-polarized” also recoded the original data to a binary trait but additionally excluded all accessions with labels located in the middle of the resistance spectrum (“susceptible”, “medium resistance”, and “resistance” classes). Furthermore, accessions in the “highly susceptible” class which had a downy mildew incidence percentage lower than the highest incidence value recorded for an accession in the “susceptible” class were also removed.

The optimal hyperparameters identified by the grid search cross-validation for the RF model trained on the “original” data were: {class_weight: “balanced”, n_estimators: 4000, max_depth: 3, max_features: 0.1, min_samples_split: 2, min_samples_leaf: 25, bootstrap: True, criterion: “gini”}. For the “binary-split” RF model: {class_weight: “balanced”, n_estimators: 4000, max_depth: 4, max_features: 0.2, min_samples_split: 2, min_samples_leaf: 25, bootstrap: True, criterion: “gini”}. Finally, for the RF model trained on the “binary-polarized” data: {class_weight: “balanced”, n_estimators: 400, max_depth: 3, max_features: “log2”, min_samples_split: 20, min_samples_leaf: 20, bootstrap: True, criterion: “gini”}.

To obtain unbiased benchmarks of the final model performances, the prediction results were compared with dummy classifiers trained with the “most frequent” strategy. These dummy classifiers predicted for each accession the class that occurred most frequently in the data. Basis for comparison were the prediction accuracy scores, i.e., the fraction of correctly predicted class labels for accessions in the test data.

To assess RF model feature consistency, 100 models were trained on the “binary-polarized” data with the hyperparameters described above, but using different seeds for the random state. The k -mer feature importances were then extracted from the model objects and the union and intersect of k -mer sets with non-zero feature importance were analyzed.

Principal component analysis (PCA) and linear discriminant analysis (LDA) were performed in Python utilizing the **Scikit-learn** module, on the pre-filtered set of 100,000 k -mers for the downy mildew resistance trait obtained by the approximate association testing step of the k -mer-based GWAS. Two and three-dimensional plots of the transformed data were generated by projecting the data onto the first principal components in the case of PCA, or the most discriminative directions in the case of LDA.

2.4 Computational implementation and performance

For scripting, Python and GNU bash version 4.4.20(1)-release were used. The Python code was executed using Python 3.9.16 and the following module versions: `jupyter_core` 5.1.1, `matplotlib` 3.6.2 (Hunter, 2007), `numpy` 1.23.5 (Harris et al., 2020), `pandas` 1.4.4 (The pandas development team, 2020), `polars` 0.19.2 (Vink et al., 2025), `pysam` 0.22.1 (Pysam developers), `qplot` 0.3.3 (Huang & Libiseller-Egger, 2022), `scikit-learn` 1.3.1 (Pedregosa et al., 2011), `scipy` 1.6.2 (Virtanen et al., 2020), and `seaborn` 0.12.1 (Waskom, 2021). Machine learning analyses and data visualization were performed in the Jupyter Notebook environment, version 7.3.5. All plots depicted in this study were generated using Python.

2.4.1 Computational resources

Downloading of the raw sequencing data, k -mer counting, k -mer-based GWAS, and read extraction were performed using the supercomputer resources provided by the Vienna Scientific Cluster (VSC), running under the AlmaLinux 8.8 operating system. Each node of the VSC is equipped with two 64-core AMD EPYC 7713 (Milan) CPUs with a base frequency of 2 GHz, a 2 TB NVMe disk, and between 512 GB to 2 TB memory.

Machine learning, k -mer downstream analyses, and the SNP-based GWAS were executed on a computing cluster running under CentOS 7 and AlmaLinux 8. This cluster comprised nine nodes having up to 64 cores of 3.8 GHz, and 1 TB memory.

2.4.2 Runtime evaluation

The process runtimes for the computational steps of the k -mer-based workflow can be found in Table 5 of Appendix C. The runtime for the SNV-based GWAS is also found under Appendix C. Runtimes for the machine learning part varied due to hyperparameter optimization and model training.

3 Results

The original sequencing data covered 303 spinach accessions, comprising 292 cultivated *Spinacia oleracea*, seven wild *S. turkestanica*, three wild *S. tetrandra*, and one unidentified species. Assuming a genome size of 989 Mbp for *S. oleracea* (Arumuganathan & Earle, 1991), the mean sequencing coverage was estimated to be 18.3x. Analysis of the metadata provided by Cai et al. (2021) revealed that 160 of the samples originated from Europe, 72 from Asia, 30 from North America, and four from Africa, with 37 accessions lacking geographic data. Among countries, Turkey (64 accessions), the United States (30 accessions), China (29 accessions), and Russia (24 accessions) were the most represented.

3.1 *K*-mer-based genome-wide association studies

For all subsequent analyses, 31 base pair long *k*-mers were used.

3.1.1 Data preparation and quality assessment

To get an overview of the size of the complete *k*-mer data set without filtering for a minor allele count (MAC), the *k*-mers were once counted in their canonized form, counting only *k*-mers that were encountered at least two times in that accession but without filtering for a minimum presence in a certain number of individuals. Then, *k*-mers were counted in the form in which they appeared in the sequencing reads without any threshold.

To assess the plausibility of the obtained data, the number of distinct counted *k*-mers for each accession was plotted against the estimated coverage. If saturation of *k*-mer counts had not yet been reached, it would be expected that the number of distinct *k*-mers would increase with the estimated coverage. However, this trend was not observed in the present data, as the *k*-mer counts remained within a stable range across the estimated coverages, with the exception of two accessions (Figure 4A). This suggests that the estimated coverages ranging from 14x to 26x (Figure 4B) are sufficient for reaching saturation, implying that most of the genome should be reflected in the counted *k*-mers. Figure 4 also shows that accessions with similar coverages can have different counts of distinct *k*-mers. This could be explained by heterozygosity, because each variant at a position in the genome would lead to duplicate distinct sequences of *k*-mers depicting a single stretch in the genome. The theoretically possible number of *k*-mers counted from a single position can be calculated by taking two to the power of the number of heterozygous alleles in the *k*-mer sequence, meaning that high heterozygosity can strongly

increase k -mer numbers. The two outliers, with sample names US355 (SRR15321797) and US356 (SRR15321665), both originated from Georgia, Europe, and were of the species *S. tetrandra*. A possible explanation could be sequencing errors or contamination, considering that they originated from the same country. As a consequence of sequencing foreign genomic material or introducing atypical variation through sequencing errors, the number of distinct k -mers counted would increase.

The counted canonized k -mers were then combined to a single table. Only those k -mers were retained which appeared in both their canonized and non-canonized form in at least 20 percent of accessions. This complete data set comprised 4,453,174,081 k -mers across 303 accessions. Most of the k -mers appeared only in a very low number of accessions (Figure 5). To reduce the size of the resulting data set, the final table storing the presence/absence of k -mers was generated by filtering for a minor allele count of 5, which corresponds to a minor allele frequency (MAF) of around 1.5 %. This reduced the number of k -mers down to 858,232,023, filtering out about 81.75 % of very rare k -mers (Figure 6).

To confirm the assertion that the spinach raw sequence reads adequately covered the genome for k -mer count saturation, four accessions were selected, two with high and two with low coverages. The number of sequencing reads was then downsampled to fixed sizes (40 M, 80 M, 100 M, and the original size) and the canonized k -mers were counted again for each reduced set. Figure 7 shows these counts of distinct k -mers plotted against the downsampled and original sequencing read numbers. Linear extrapolations between the lowest two numbers of sequencing reads demonstrate the expected increase in counted k -mers if saturation is not yet reached. However, the actual number of k -mers were below the expected values. When more than tripling the number of sequencing reads, k -mer numbers increased only by roughly one fifth to one fourth; the weaker slope at higher numbers suggests k -mer count saturation.

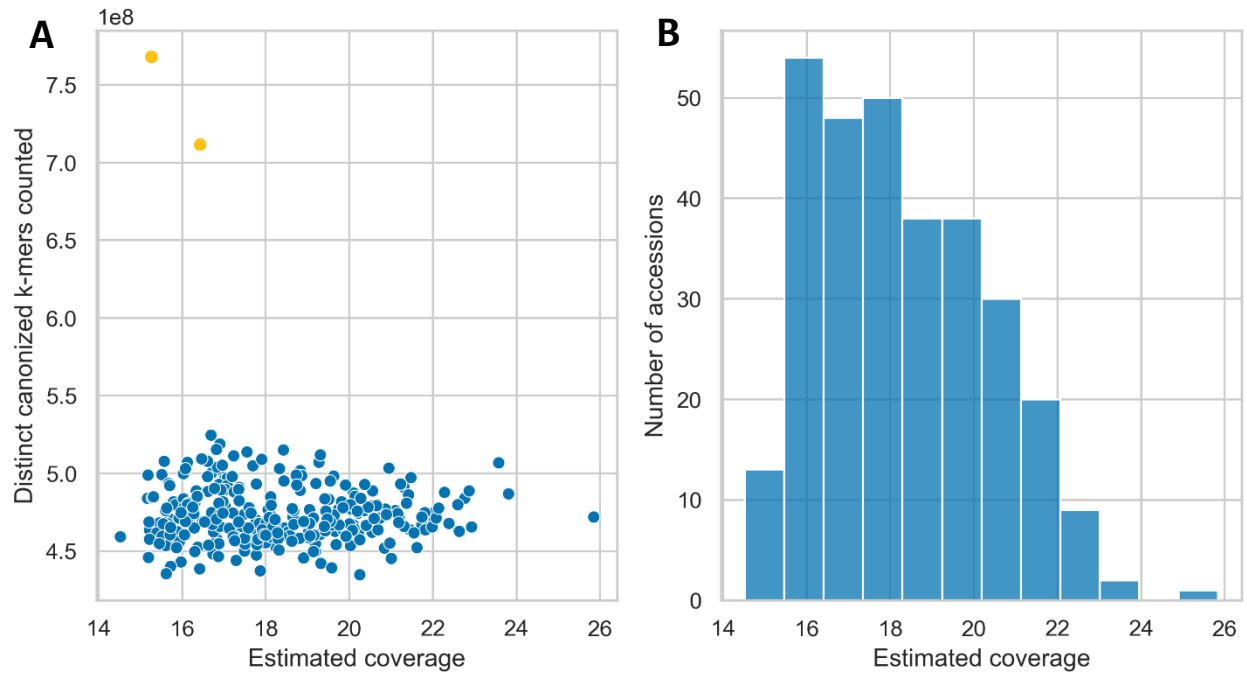


Figure 4: K -mer counts plotted against the coverage. (A) The number of distinct canonized k -mers counted in each accession plotted against the estimated coverage. Two accessions (highlighted yellow) exhibited unusually high k -mer counts. (B) Histogram of the estimated coverages of the 303 spinach accessions.

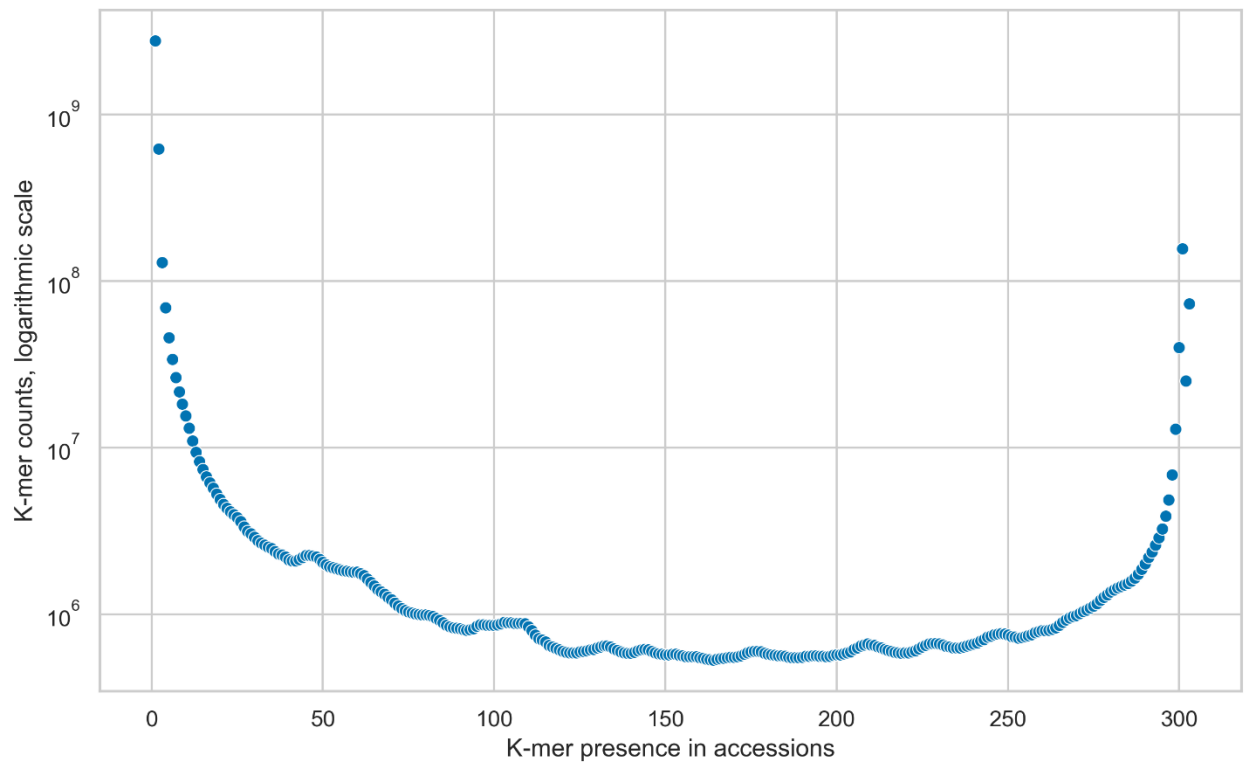


Figure 5: Distribution of k -mer counts across individuals. Logarithmic-scale number of distinct k -mers counted with a presence in x accessions. Most k -mers were either present in very few or most of the accessions, with over 1 billion k -mers being only counted in single accessions.

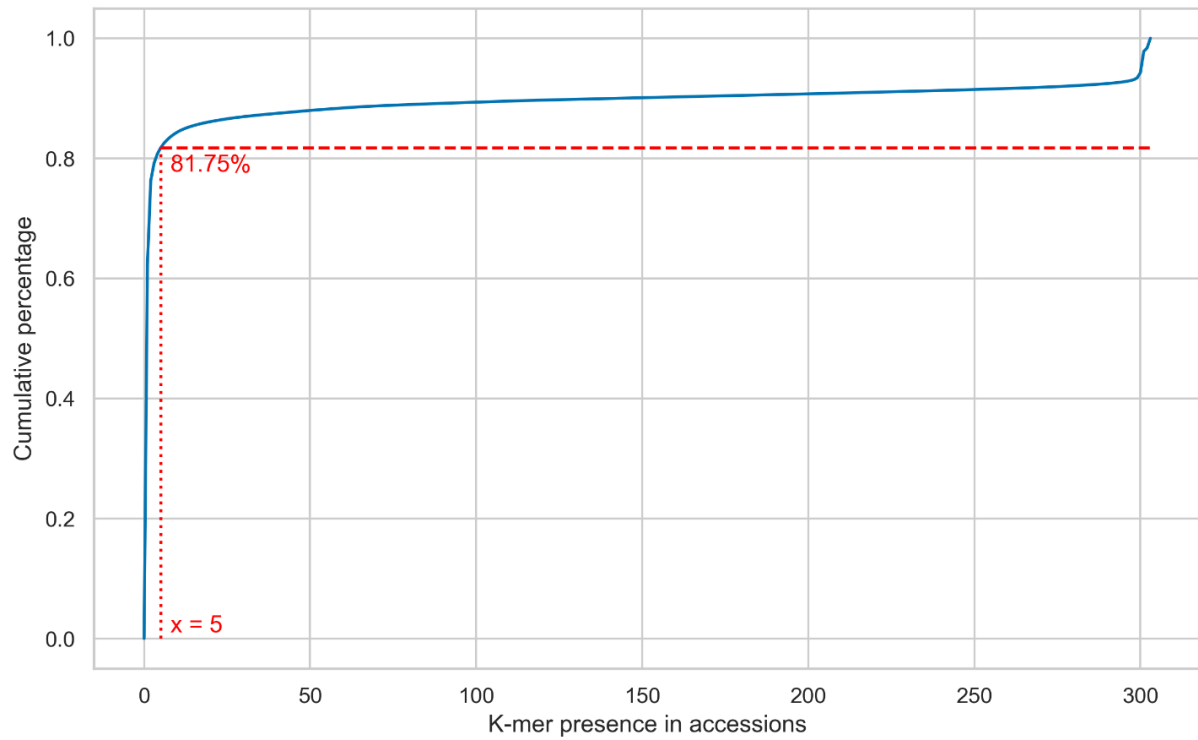


Figure 6: Cumulative percentage of distinct k -mer presence in accessions. The dotted red line shows the minor allele count of five used for filtering the k -mer table. Around 81.75 % of k -mers were present in five accessions or less.

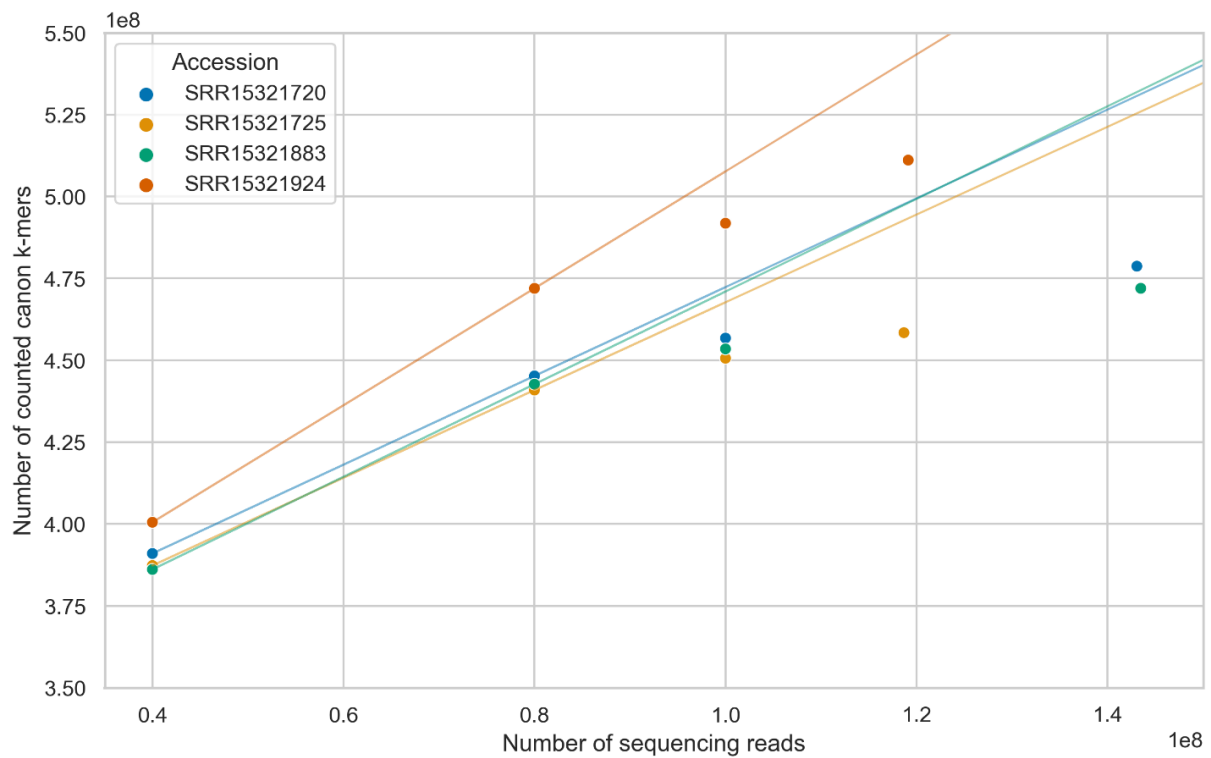


Figure 7: Number of canonized k -mers counted for four accessions at different numbers of (downsampled) sequencing reads. The lines are extrapolations of the increase of counted k -mers between the two lowest numbers of downsampled reads.

3.1.2 Genome-wide association analysis using k -mers

Before running the association tests, a kinship matrix was calculated based on the k -mer presence/absence, visualizing the biological relatedness between pairs of individuals. A value of one would indicate that the k -mers counted were identical between two accessions, while a relatedness of zero implies no overlap in the found k -mers. The mean relatedness without self-comparisons was 0.81, the lowest relatedness 0.63, and the maximum relatedness was 0.93 between the accessions WS242 (SRR15321800) and WS245 (SRR15321799). The hierarchically clustered kinship matrix showed clear differences in relatedness among spinach accessions, suggesting the presence of underlying population structure (Figure 8). The two accessions US355 (SRR15321797) and US356 (SRR15321665), which were outliers regarding their unusually high k -mer counts (Figure 4) also clustered separately from other accessions in the kinship matrix (Figure 8, positioned in the leftmost two columns or first two rows). While the relatedness of 0.92 between them is high, their similarity to the other accessions is relatively low. This supports the notion of common contamination or sequencing errors affecting these two accessions.

The kinship matrix was used to correct for population structure when running the k -mer-based genome-wide association study (GWAS) with GEMMA (X. Zhou & Stephens, 2012) as part of the *kmersGWAS* pipeline (Voichek & Weigel, 2020). A total of 20 agronomic traits of spinach previously studied by Cai et al. (2021) were examined in the GWAS. These included traits concerned with plant and leaf morphology, petiole characteristics, bolting and flowering times, sex type, oxalate content, and resistance to a fungal pathogen causing the downy mildew (DM) disease. Of the 20 traits measured, eleven were continuous variables and the remaining nine were categorical (discrete). **Table 1** lists the categories used for the discrete traits and the ranges observed for the continuous traits.

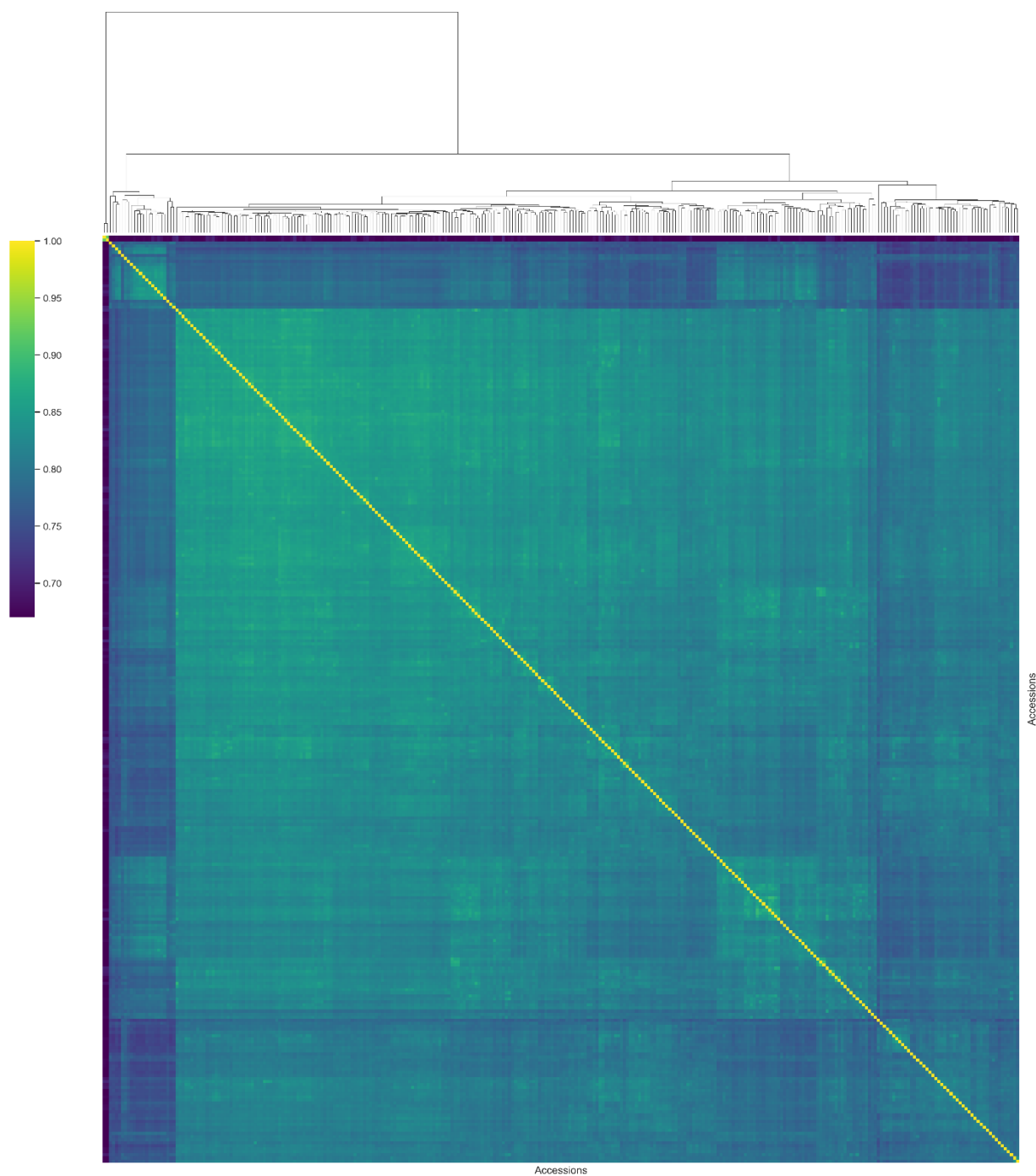


Figure 8: Hierarchically clustered kinship matrix based on the presence/absence of 858,232,023 k -mers. Illustrating the pairwise relatedness between the 303 accessions, where higher values indicate greater identity regarding the counted k -mers. The color scale represents the genetic similarity between samples, with lighter green shades indicating higher proportions of shared k -mers and darker blue shades representing fewer shared k -mers.

Table 1: List of all 20 agronomic traits investigated in the genome-wide association studies. Taken from the study by Cai et al. (2021). The second column shows the classes with the counts of accessions for discrete traits or the range of values in the case of continuous traits.

Agronomic trait	Classes/range	Missing values
Plant type	Erect (1), semi-erect (33), spreading (264)	5
Plant height	1.7–31.5	6
Plant width	10.63–60.97	6
Leaf length	3.67–22.13	6
Leaf width	1.6–13.5	6
Leaf number	7–34	5
Leaf surface texture	Wrinkled (234), smooth (64)	5
Leaf shape	Orbicular (18), ovate (86), elliptic (193), halberd (1)	5
Leaf apex shape	Non-acute (264), acute (34)	5
Leaf base shape	Non-plane (266), plane (32)	5
Leaf division	None (92), shallow (108), medium (52), deep (46)	5
Petiole length	1.53–19.87	6
Petiole width	0.1–1.1	6
Petiole color	Green (240), purple (58)	5
Bolting days	40–203	1
Flowering days	93–230	1
Sex type	Female (140), male (52)	111
Oxalate content	0.59–1.93	11
DM resistance	Highly susceptible (196), susceptible (62), medium resistance (13), resistance (1), high resistance (23)	8
DM incidence	0–100	8

To further reduce the size of the GWAS input data and decrease computational demands, a MAF of 5 % (corresponding to a MAC of 16) was used in the subsequent steps of the GWAS.

In addition to filtering by the minor allele frequency, an initial approximate association testing was performed to reduce the size of the dataset and retain only the 100,000 top-scoring *k*-mers.

The exact association testing was done with linear mixed models. The pre-filtered 100,000 *k*-mers from the previous step were used as input to get exact significance scores. When performing GWAS, multiple hypothesis testing is a major issue. To control the type I error rate

(i.e., false positives) without losing too much statistical power, the phenotypes were randomly permuted 100 times and a genome-wide association test was conducted on each permutation. To obtain significance level thresholds for the 5 % and 10 % family-wise error rates (FWER), the lowest p -value from the $n\%$ -quantile of the permutation test distribution was taken. Because this step was computationally demanding—requiring 100 GWA tests for each trait—the preceding step of reducing the k -mer table from over 800 million to 100,000, as previously described, was indispensable. The p -value thresholds obtained this way ranged from $10^{-8.83}$ to $10^{-10.09}$.

Table 2: Number of significant k -mers found in the k -mer based GWAS at the 5 % and 10 % family-wise error rate (FWER) levels. For the 5 %, the number of genes with at least one mapped sequencing read extracted from the k -mers are listed. No significant k -mers were found for the four traits leaf division, leaf width, plant height, and plant width.

Agronomic trait	Number of significant k -mers		Identified genes
	5 % FWER	10 % FWER	5 % FWER
Plant type	135	164	128
DM resistance	107	158	79
Petiole Width	58	79	68
Leaf apex shape	48	53	61
Leaf number	29	46	31
Leaf base shape	28	39	68
Leaf surface texture	17	23	3
Petiole color	11	17	10
Bolting days	15	16	2
Petiole length	6	11	0
Oxalate content	0	4	0
Sex type	1	4	11
Flowering days	1	2	25
Leaf shape	1	2	3
DM incidence	0	1	0
Leaf length	0	1	0

Association tests were then conducted on the non-permuted phenotypes, and likelihood ratio test p -values were calculated for each k -mer. A k -mer was considered significantly associated with a phenotype if its p -value was below the respective significance threshold. Table 2 lists the numbers of significant k -mers at the 5 % and 10 % FWER. For all further analyses, only the k -mers passing the 5 % FWER threshold were used, limiting the extent of type I errors and to allow to focus on the most likely relevant variants.

Downstream analyses of the k -mer-based GWAS results consisted of first extracting all the raw sequencing read pairs containing the significant k -mers and their reverse complements. These reads were then mapped to the spinach reference genome. Important genic regions were identified by finding overlaps between the genome annotation and regions with mapped reads or “hits” (Table 2). To estimate the importance of a gene, they were sorted by the number of extracted reads mapping to that specific region. In general, a higher “hit” count should signal a higher importance because it was confirmed by a larger number of reads. Because this method is only approximate and strongly influenced by sequencing depth and k -mer frequency, it cannot be used to sharply define important genes. It must be interpreted for each trait individually, where it can serve to distinguish genes strongly supported by mapped reads from genes with fewer extracted reads mapped to them. It was necessary to find such a metric because a single k -mer often mapped to multiple positions in a genome. Additionally, each significant k -mer had to be studied both in its canonical and reverse complement form, as the strand information had been lost when counting the k -mers from the raw sequencing reads. This further increased the number of extracted reads and the likelihood of them mapping to different positions in the reference genome. The rationale behind using the count of mapped reads as a measure of a gene’s importance is that while some genes will be randomly matched by a few reads, the consensus across all studied accessions will highlight the truly correlated genes. The trait “plant type” had the highest number of identified genes (128), followed by “downy mildew resistance” (79, Table 2). Hereafter, only the downstream analysis for the downy mildew resistance trait will be presented in detail.

3.1.3 Identification of candidate genes for downy mildew resistance

The downy mildew (DM) resistance trait in spinach is of particular interest to plant breeding companies and researchers. Cai et al. (2021) identified genes strongly associated with the resistant phenotype, making their findings a valuable resource for validating the effectiveness of the k -mer based GWAS. Consequently, the results for DM resistance were compared to those of Cai et al. (2021) and used to validate the effectiveness of this approach.

A total of 3759 sequencing read pairs, containing any of the 107 *k*-mers or their reverse complements significantly associated with downy mildew resistance, were extracted from the 303 spinach accession raw read datasets. Of these reads, 96.24 % were mapped to the reference genome, with 31.65 % mapping to positions within annotated regions. In total, 79 distinct genes were matched to at least one sequencing read. There were 2939 read pairs (78.19 %) that mapped to more than one position in the genome.

The extracted reads mapping inside annotated regions were found predominantly around the start of chromosome 3. Of the extracted reads, 712 mapped to a gene identified as *SOV3g001250* in the Monoe-Viroflay spinach genome annotation (Cai et al., 2021). This gene, spanning positions 1,246,993 to 1,253,083, is predicted to code for a putative disease resistance protein. Two nearby genes, *SOV3g000990* and *SOV3g001240*, were also well-supported by 626 and 209 mapped reads, respectively. They were annotated as a disease resistance protein and as an unknown protein. Another gene, *SOV2g037000* coding for a Putative O-acyltransferase WSD1-like, is still relatively well-supported by 132 sequencing reads. The WSD1 enzyme plays a key role in wax ester synthesis which catalyzes the biosynthesis of cuticular waxes and is mainly expressed in stems, leaves, and flowers (F. Li et al., 2008). All other annotated genes had fewer than a hundred reads mapping to them. Although the method did not allow defining an exact threshold for identifying significant genes, genes with a low number of mapped reads were deemed insufficient for drawing reliable conclusions and were therefore not considered strongly associated.

As mentioned previously, around two-thirds of the extracted reads did not align within an annotated region. These signals were more difficult to interpret in a systematic way. As a novel method of visualizing the results obtained from a *k*-mer-based GWAS, a type of plot has been developed that charts the extracted sequencing reads against the positions on the reference genome to which they mapped. Due to their functional similarity to the Manhattan plots commonly used in traditional GWAS, they can be referred to as *k*-mer-based Manhattan plots. In contrast to regular Manhattan plots, these plots do not show the *p*-values of SNPs. Instead, the negative logarithmic *p*-value of the significant *k*-mer contained in the read (or the lowest in case of multiple *k*-mers) is plotted on the *y*-axis. To highlight those positions which are supported by multiple reads, the data points were plotted with 95 % transparency (a position was plotted with full opacity if at least 20 reads were mapped to it). This is important to discriminate spurious associations arising from multi-mapping sequencing reads and reads that were coincidentally extracted due to the reverse complements of significant *k*-mers (Appendix D).

The k -mer-based Manhattan plot for the DM resistance showed a strong signal at the start of chromosome 3, which was supported by a large number of mapped reads (Figure 9). Additionally, there were well-supported signals at the start of chromosome 2 and in chromosomes 4 and 5. Some extracted reads mapped to contigs that were not assigned to any specific chromosomes; these contigs were grouped together under the label “SOVchro”. When there are several faint data points ordered along a horizontal line, they likely originated because of multi-mapping sequencing reads stemming from the same k -mer. In contrast to conventional Manhattan plots, only the reads extracted from the significant k -mers are shown in the k -mer-based Manhattan plots.

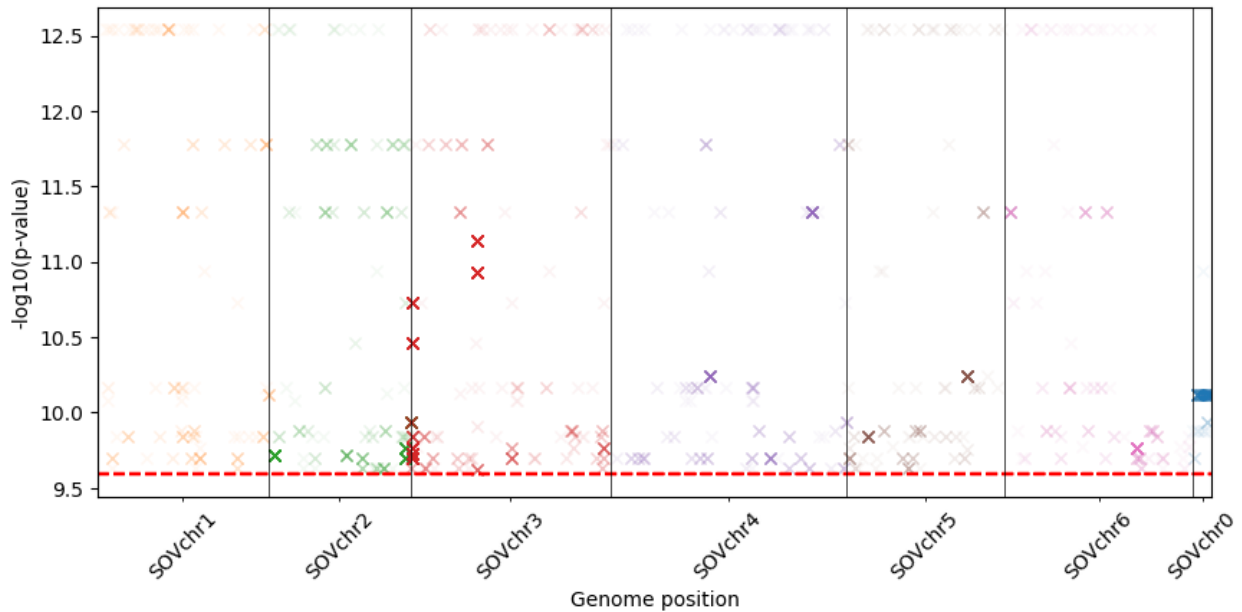


Figure 9: K-mer-based Manhattan plot of GWAS results. Displayed are the lowest p -values of significant k -mers contained in extracted sequencing reads and their corresponding genomic mapping positions, including unanchored scaffolds (collected in “SOVchro”). The dashed red line indicates the genome-wide significance threshold of $10^{-9.59}$ for a 5% family-wise error rate. The higher the opacity of a data point, the higher the number of supporting mapped reads.

To examine promising results from the k -mer-based GWAS related to downy mildew resistance, the reads containing significant k -mers were extracted from the raw sequencing data and mapped to the reference genome. These mapped reads were then analyzed in detail using a genome browser. The putative resistance gene *SOV3g001250* is a well-known candidate gene for DM resistance and therefore serves as a good reference for comparing the results of the k -mer-based GWAS (Bhattarai et al., 2022; Cai et al., 2021). When inspecting the gene, two clusters of reads were found: one at the start of the gene and the other one around the center (Figure 10). By aligning the mapped reads with the reference genome, it was possible to identify the genetic variation captured by the reads. A table with SNPs detected inside mapped reads containing

significant k -mers can be found in Appendix A. The following eight k -mers were found to be significantly associated with this gene:

- TGTATTGACGGTCAGGAACGTTCTCATTTCAG
- GTATTGACGGTCAGGAACGTTCTCATTTCAGG
- AATCTGTTCCAATTATAGAATAGTTGAAGGC
- ATCTGTTCCAATTATAGAATAGTTGAAGGCA
- TCTGTTCCAATTATAGAATAGTTGAAGGCAT
- CTGTTCCAATTATAGAATAGTTGAAGGCATG
- TGTTCCAATTATAGAATAGTTGAAGGCATGA
- GTTCCAATTATAGAATAGTTGAAGGCATGAA

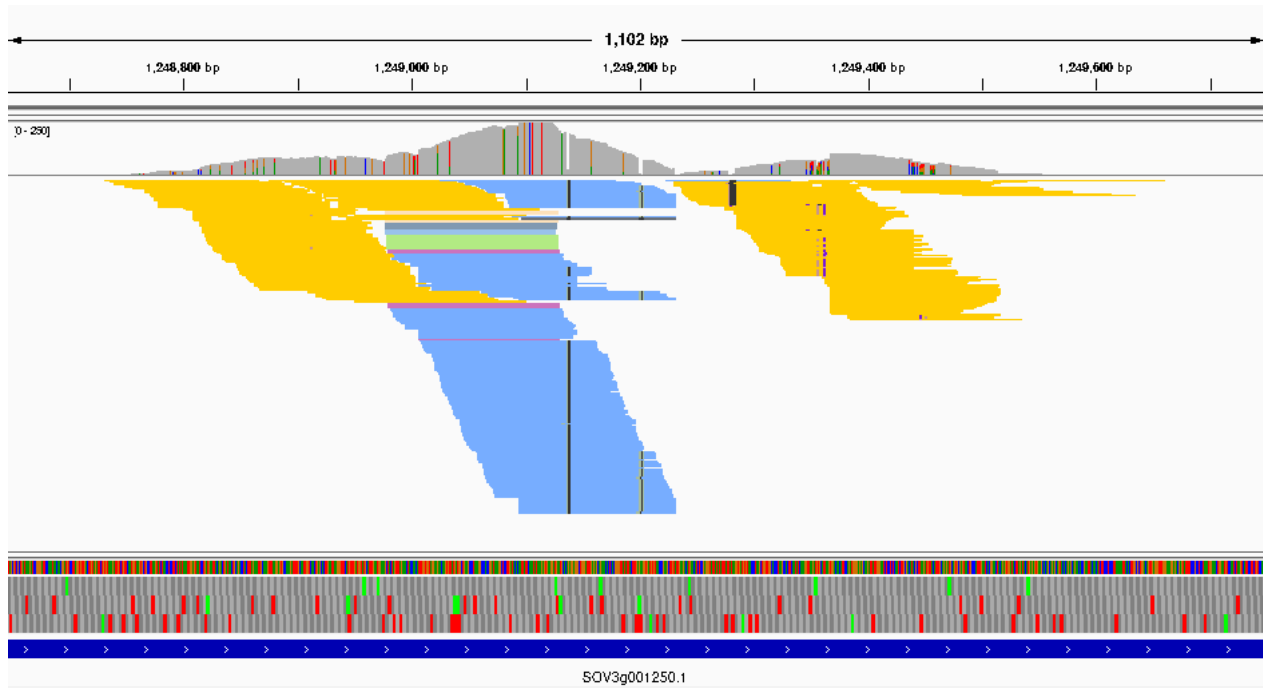


Figure 10: Detailed view in the genome browser IGV of mapped sequencing reads. Shown are reads containing k -mers significantly associated with DM resistance, mapped to the putative disease resistance gene *SOV3g001250*, located on chromosome 3. Reads are colored based on the k -mers they contain; yellow reads did not contain k -mers but are mates of the k -mer-containing reads.

To visualize the presence/absence patterns of the significant k -mers, hierarchically clustered heatmaps were generated for each phenotypic trait. Accessions were ordered by their respective phenotype. By grouping k -mers according to their presence or absence across the samples in a heatmap of the 107 k -mers significantly associated with downy mildew resistance, patterns of genetic variation emerged (Figure 11). As expected, these k -mers were found to be more frequently present in resistant accessions. Moreover, some groups of k -mers were either nearly all present or all absent from a specific accession. These patterns of co-occurrence are likely due to the fact that a single site of genetic variation can be covered by multiple k -mers. Different presence/absence patterns in resistant accessions indicate that downy mildew resistance has a

complex genetic architecture with multiple resistance mechanisms and a diverse genetic background. The heatmaps also aid in identifying specific k -mers that are consistently associated with particular traits, making them potential markers for marker-assisted selection (Thudi et al., 2021).

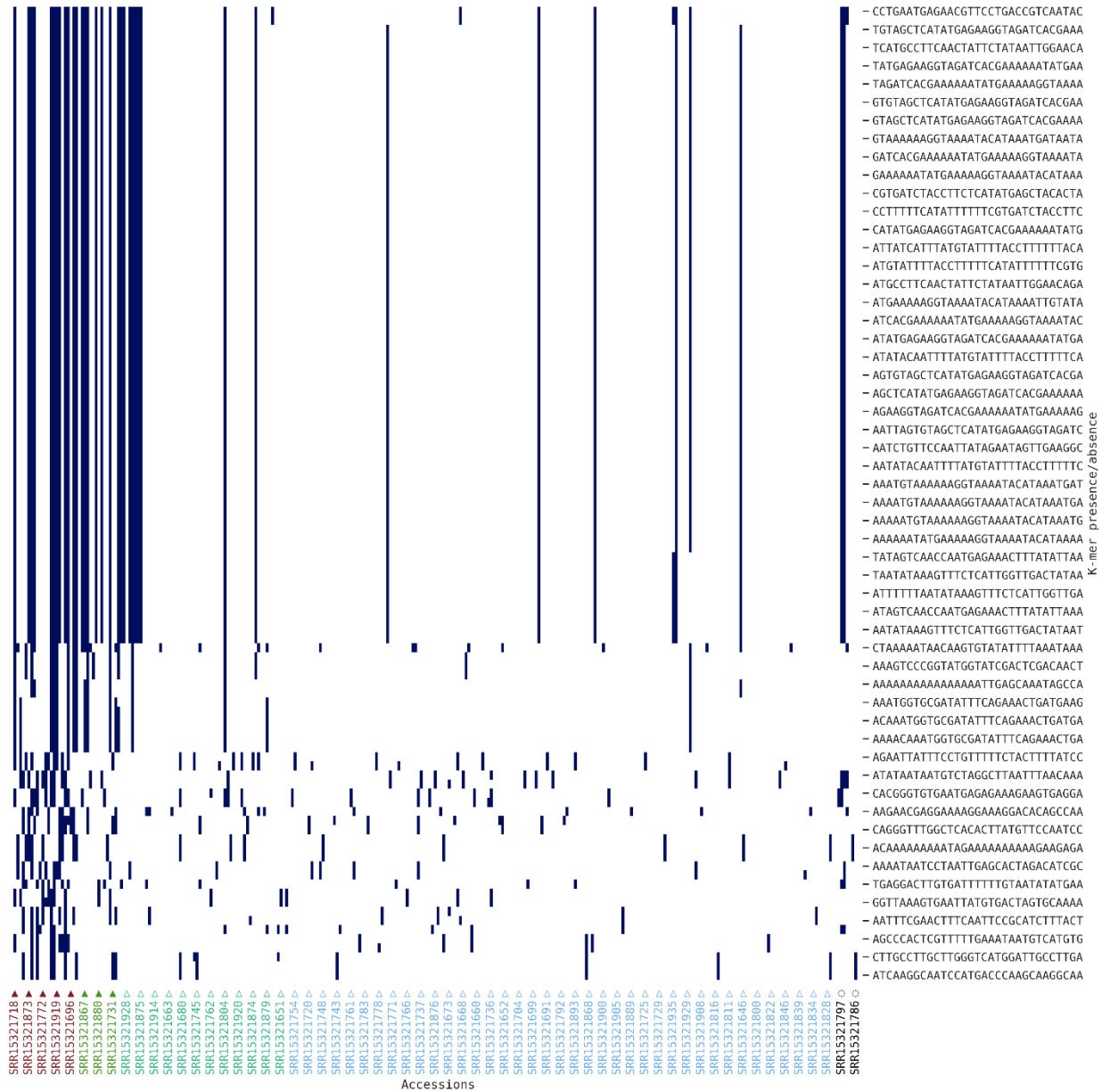


Figure 11: Hierarchically clustered presence/absence patterns of the 107 k-mers significantly associated with downy mildew resistance. Accession labels are ordered from left to right and colored according to their resistance phenotype: high resistance (dark red), medium resistance (green), susceptible (turquoise), highly susceptible (light blue); black indicates missing phenotypic data. Solid triangles denote resistant phenotypes, hollow triangles susceptible ones. Not all accession and k -mer labels are shown.

3.1.4 Computational performance of *k*-mer-based GWAS

Examining the runtimes (Appendix C), it is evident that *k*-mer counting represents the most computationally demanding step unique to the *k*-mer-based workflow. Notably, downstream analyses of *k*-mers took longer than the association tests themselves. This increased processing demand arose because, for each trait, the entire set of raw sequencing reads, totaling several terabytes, had to be searched for significant *k*-mers and their reverse complements. A specialized tool developed as part of this study considerably reduced execution times for this step compared to a naïve search, while also providing tailored match statistics.

3.2 SNV-based genome-wide association studies

To compare the results of the *k*-mer-based GWAS with a traditional variant-based approach, additional genome-wide association studies were conducted using single nucleotide polymorphisms (SNPs) and small insertions/deletions (indels). These variants will be referred to as "SNVs" (single nucleotide variation). The Variant Call Format file containing the called and filtered variants was obtained from a previous study by a colleague, An Nguyen-Hoang, M.Sc. (Nguyen-Hoang et al., 2024).

To assess the quality of the variants, summary statistics and a quality control report were generated. The VCF file contained 4,580,829 variants across 305 samples. Of those, 4,166,093 were SNPs and 414,736 were indels. The ratio of transitions to transversions (Ti/Tv) was 1.34. The maximum length of called indels was 60 bp.

Additional QC steps were performed using the PLINK toolset. Five commonly used QC metrics (Marees et al., 2018) were investigated in detail: (i) SNP missingness, (ii) individual missingness, (iii) minor allele frequencies, (iv) heterozygosity, (v) Hardy-Weinberg equilibrium. From the original 4,580,829 variants, a total of 367,751 were removed due to missing genotype data, 601,582 failed the Hardy-Weinberg exact test, and 1,376 were filtered out due to minor allele frequency thresholds. Among the 305 samples, two were excluded due to missing genotype data. After filtering, the final dataset used for GWAS contained 3,610,120 variants across 303 accessions.

3.2.1 Genome-wide association analysis using SNVs

To analyze sample kinship and correct for population structure in the SNV-based association tests, a centered relatedness matrix was estimated using GEMMA (Figure 12). This genetic relationship matrix is mean-centered, meaning that column means are subtracted from each

column. Consequently, the diagonal values are not necessarily equal to 1. Negative values indicate that a pair of samples share fewer alleles than the population average, while positive values denote greater relatedness between accessions.

Genome-wide association tests were then conducted for the 20 phenotypic traits listed in Table 2, using linear mixed models. To account for population stratification, a centered relatedness matrix was used in the LMM calculations. For each SNV, a p -value was calculated. To correct for multiple testing, a Bonferroni-adjusted p -value threshold of $1.38 \cdot 10^{-8}$ was used. The numbers of significant variants obtained for each trait using this adjusted significance cutoff are listed in **Table 3**.

Table 3: Number of significant SNPs, indels and genes affected by those variants found in the SNV-GWAS. Significant SNVs were selected using the Bonferroni-corrected p -value threshold of $1.38 \cdot 10^{-8}$. No significant SNVs were found for the six traits plant width, plant height, oxalate content, leaf shape, leaf length, and leaf division. Genes were counted as affected if the variant lies within the genic region or 5 kb upstream or downstream.

Agronomic trait	Number of significant variants				Number of affected genes
	SNPs	Insertions	Deletions	Total	
Leaf number	198	10	15	223	79
DM resistance	142	9	7	158	98
Plant type	84	1	7	92	32
Petiole color	62	0	1	63	7
Bolting days	33	0	4	37	13
Leaf surface texture	8	0	1	9	5
Petiole length	7	0	1	8	2
Leaf base shape	4	0	1	5	4
Leaf width	3	0	0	3	1
Sex type	1	1	0	2	0
Leaf apex shape	1	1	0	2	1
Petiole width	1	0	0	1	0
Flowering days	1	0	0	1	1
DM incidence	1	0	0	1	1

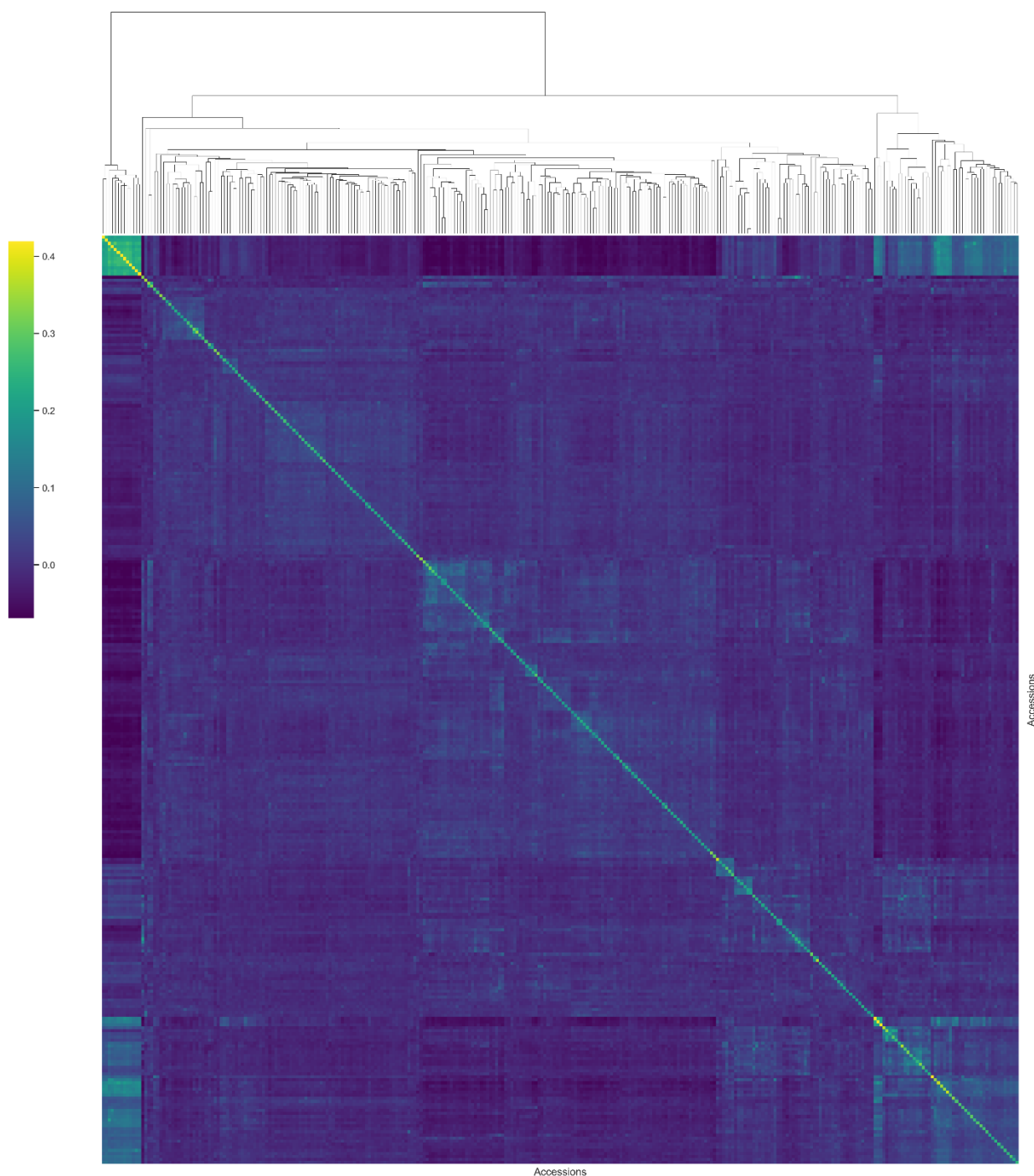


Figure 12: Hierarchically clustered heatmap of the centered pairwise relatedness matrix between the 303 accessions. Calculated using the genetic covariance between individuals based on 3,610,120 SNVs. Each cell corresponds to the pairwise relatedness between two accessions (listed along the axes), with lighter (green) tones indicating greater relatedness and darker (blue/purple) tones indicating lower relatedness. The dendrogram on the top shows the hierarchical clustering of samples.

Manhattan plots were generated for each trait by plotting the negative logarithmic p -values of each variant against its location on the genome. In addition to the genome-wide significance threshold of $1.38 \cdot 10^{-8}$, a suggestive threshold of $2.77 \cdot 10^{-7}$ (for an unadjusted alpha level of 1) was calculated for these plots (Appendix E). Additionally, the significant variants obtained for each trait were extracted into separate VCF files and analyzed regarding their type, location, functional effect, and impact. The prediction and annotation of these genetic variants enabled the identification of variant types (SNP, insertion, or deletion) and their potential influence on genes. This analysis included counting the number of genes potentially affected by variants located within genic regions or within 5 kb upstream or downstream (Table 3), as well as evaluating their biological impact.

3.2.2 Genetic variants associated with downy mildew resistance

Because of its agronomical relevance, the downstream analysis of significant variants again focused on the downy mildew resistance trait. Functional effect prediction and genomic variant annotation of the 158 identified variants revealed that 142 of them were SNPs, 9 were insertions, and 7 were deletions. By far the most variants were located on chromosome 3 (143), while the others were found on chromosomes 1, 2, 5, 6, and on an unassigned scaffold.

In total, 312 variant effects were detected. 9.0 % of effects had a low putative impact on their gene, 4.5 % had a moderate putative effect, and 86.5 % were annotated with having a modifying impact. These impact classes each encompass different terms in the Sequence Ontology (Eilbeck et al., 2005). The most common effect types were downstream and upstream gene variants, accounting for nearly half of all identified effects. Around a fifth of the effects were linked to intergenic regions. 13.5 % of effects were found in exons, and 13 % in introns. The remaining effects were located in 3' and 5' untranslated regions (UTRs). Of the 42 exonic variants, 14 led to a missense mutation (i.e., they changed one or more bases, resulting in a different amino acid sequence), while 28 were silent.

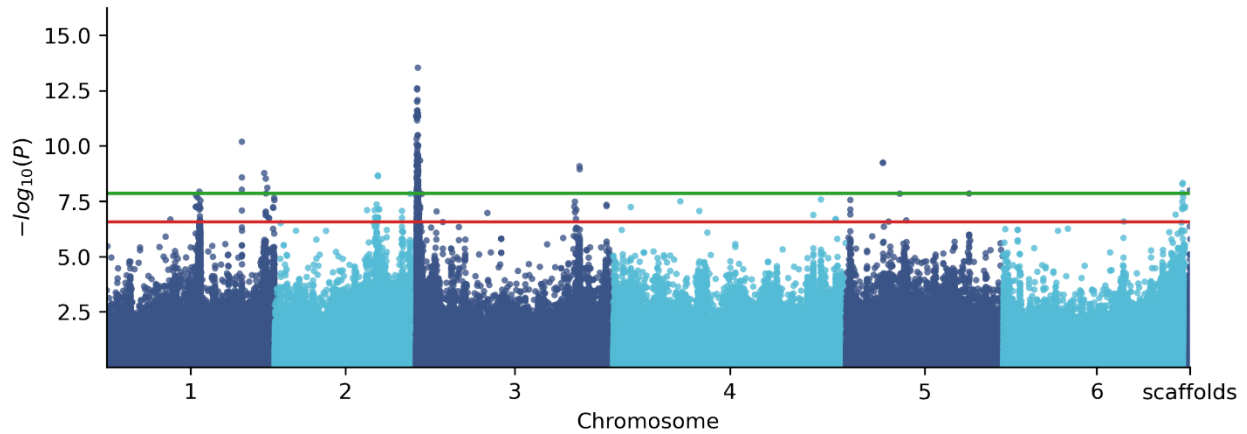


Figure 13: Manhattan plot displaying the results of the GWAS for downy mildew resistance. The x-axis represents genomic positions across six chromosomes and unanchored scaffolds, while the y-axis shows the negative logarithmic p -values of SNV-trait associations. The red and green horizontal lines indicate the suggestive and genome-wide significance thresholds (at $\alpha = 1$ and $\alpha = 0.05$, respectively), with variants above this threshold considered significantly associated with downy mildew resistance. Peaks of association suggest potential loci involved in resistance mechanisms.

To illustrate the positions of SNPs and indels significantly associated with downy mildew resistance, a Manhattan plot was generated (Figure 13). A total of 98 genes had variants located either in genic regions or upstream/downstream of them. The greatest peak was found at the start of chromosome 3, spanning from gene *SOV3g000060* to *SOV3g0001920* and approximately ranging from positions 163,182 to 2,024,852 on that chromosome. There were six (putative) disease resistance genes located inside this region: *SOV3g0000690*, *SOV3g0000710*, *SOV3g0000980*, *SOV3g0000990*, *SOV3g0001030*, and *SOV3g0001250*. Few additional variants inside genes on chromosomes 1, 2, 5, and 6 were identified, with some of them having functional annotations that were indirectly linked to disease resistance. Close to a major disease protein is *SOV3g0001280*, encoding a cysteine and histidine-rich domain-containing protein RAR1 (Muskett et al., 2002). Another nearby gene, *SOV3g0000770*, translates into a NAC domain-containing protein. Two genes within this region, *SOV3g0000280* and *SOV3g0000380*, express cellulose synthases that influence cell wall composition.

3.3 Machine learning analyses

To contrast the methodology of the k -mer-based GWAS, a machine learning (ML) approach was applied to the k -mer data. The objective was to test the hypothesis that machine learning can predict phenotypes using the presence or absence of k -mers as input features. Additionally, the goal was to determine whether insights from the model metrics could identify which k -mers are

most valuable for predicting the phenotype and to subsequently find associations between those k -mers and the phenotype.

3.3.1 Phenotype prediction using random forest models

Among the many ML classification algorithms available, a random forest (RF) classifier was chosen. RF models are robust against bias, effectively perform with high-dimensional data such as genetic variants and provide performance metrics to evaluate input features (Salman et al., 2024). Because this was intended as an initial proof-of-concept for combining ML and k -mer data and given that training ML models is computationally intensive and time-consuming, only the downy mildew resistance phenotype data was used in all ML analyses. ML models have been reported to perform better when feature selection is performed (Iranzad & Liu, 2024). To decrease the number of irrelevant or redundant features, keep the model complexity and training times low, and simplify the interpretation of the model metrics, only a reduced set of k -mers was used as input data. Specifically, the pre-filtered set of 100,000 k -mers, obtained by selecting only the top scoring k -mers from the k -mer-based approximate association testing, was chosen for this purpose.

As mentioned in Table 1, the downy mildew resistance values were very unevenly distributed across the samples. 196 accessions were classified as “highly susceptible”, 62 as “susceptible”, 13 as “medium resistance”, 1 as “resistance”, and 23 as “high resistance”. Eight accessions lacked phenotypic data. For all subsequent analyses, the data was randomly split into a training set comprising 75 % (220 accessions) and a test set comprising 25 % (74 accessions). Stratified sampling was used to account for the imbalanced phenotypes and to retain the distribution in both the training and test sets.

During training of all models, grid search in combination with cross-validation was used to tune the hyperparameters. For each possible combination of parameter values, the training set was split into five even folds. Models were trained five times, each time using a different fold as a validation set to fit and evaluate the hyperparameters. Stratified sampling was employed to ensure that the class distribution was preserved in each fold. The average accuracy score across all cross-validation folds was used to determine the best parameter combination. To compensate for the skewed effects of the uneven classes in the training data, all models were trained using “balanced” class weights.

To make meaningful comparisons of the prediction results of the machine learning models, a dummy classifier was also trained on the same data using the “most frequent” strategy. These

dummy classifiers always predicted the label occurring with the highest frequency for all samples and thus gave a benchmark that is not biased by the imbalanced phenotypes.

3.3.2 Input data transformation and phenotype prediction

In the initial attempt, the phenotype values were used as they were provided by Cai et al. (2021), except for dropping the single accession labelled with “resistance”. The dummy classifier achieved a prediction accuracy of 66 %. The model with the best parameters, determined through grid search cross-validation, achieved a prediction accuracy of 74.3 %. While all accessions in the majority class “highly susceptible” got predicted correctly, most samples in the “susceptible” class were incorrectly predicted as “highly susceptible”. Moreover, the resistant accessions were largely misclassified (Figure 14).

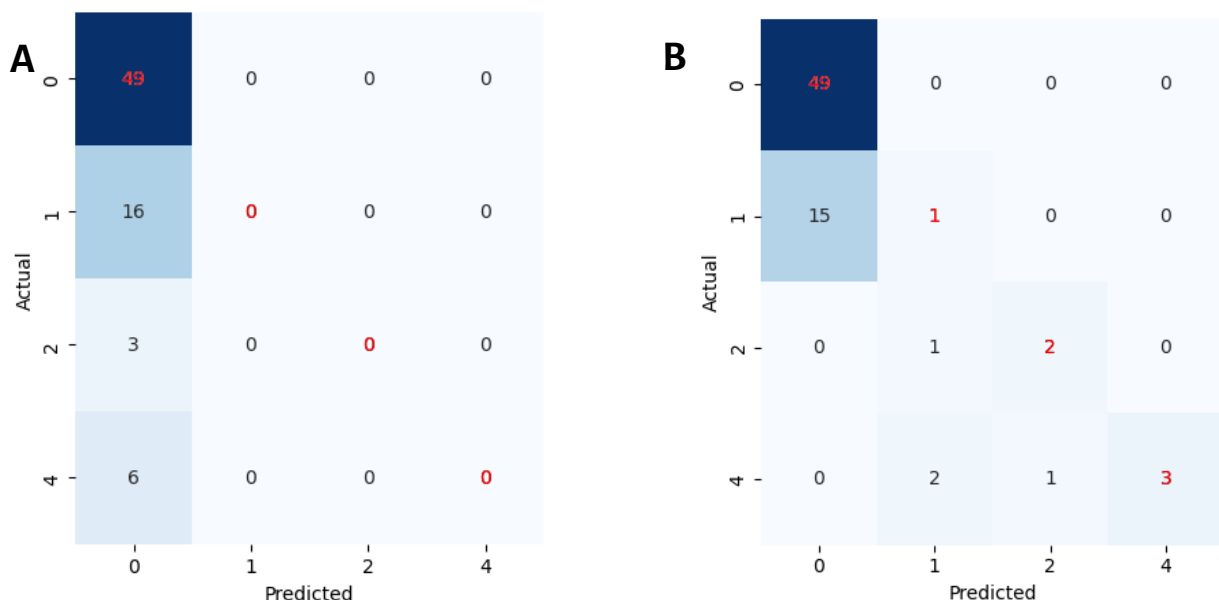


Figure 14: Confusion matrix of the ground truth and predicted resistance classes. (A) Predictions by the dummy classifier. (B) Predictions by the random forest model, trained on the original data. The labels represent the following classes: 0 – highly susceptible, 1 – susceptible, 2 – medium resistance, 4 – high resistance.

With the aim of reducing the dataset complexity and to focus on the main objective of predicting whether an accession was resistant or susceptible to downy mildew disease, the phenotype values were recoded from their original classes into a binary trait. Specifically, the “highly susceptible” and “susceptible” accessions were pooled under the label “susceptible” (258 accessions), while the accessions classified as “medium resistance”, “resistance”, and “high resistance” were pooled under the label “resistant” (37 accessions).

Due to the imbalanced classes, the dummy classifier, using the “most frequent” strategy, achieved a prediction accuracy of 88 %. The best random forest model trained on the transformed binary phenotype data achieved a prediction accuracy of 94.5 %. When fitting the model to the test data, all 65 susceptible accessions were correctly classified. However, of the nine resistant accessions, only two were predicted correctly, while seven were misclassified.

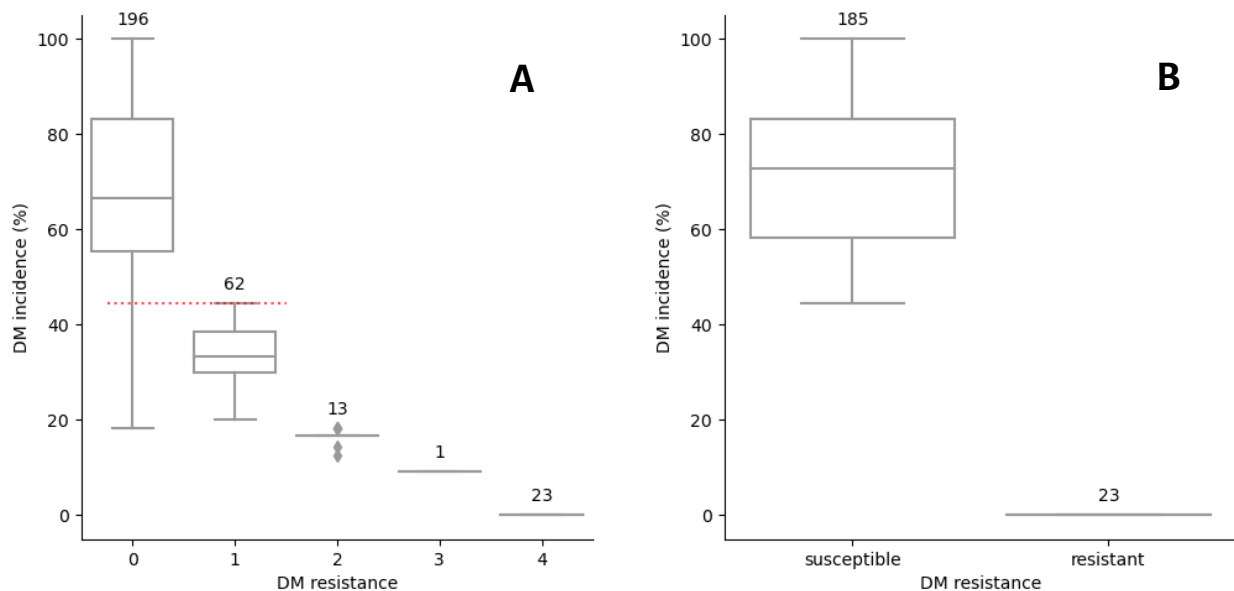


Figure 15: Boxplots of downy mildew incidence phenotype values across different DM resistance statuses. Panel (A) shows the original phenotype data, while panel (B) displays the data transformed into polarized input. The dashed line indicates the highest DM incidence in susceptible accessions, which was used as a cutoff. The labels on the x-axis represent the following classes: 0 – highly susceptible, 1 – susceptible, 2 – medium resistance, 3 – resistance, 4 – high resistance.

Plotting boxplots of downy mildew incidence values across different the different DM resistance classes revealed a non-linear relationship between the two phenotypes. More specifically, one could not be directly translated into the other, and some accessions categorized as “highly susceptible” had in fact lower downy mildew incidence percentages than samples classified as “susceptible” (Figure 15).

In another attempt to reduce dataset complexity and better capture the underlying biological distinctions, only accessions in the most extreme classes, “highly susceptible” and “high resistance,” were retained. Additionally, eleven classified as “highly susceptible” but with downy mildew incidence below the highest value in the “susceptible” class were excluded. This approach removed ambiguous samples and retained the most reliable samples with strongly marked resistance phenotypes. By using this polarized input data, the risk of including mislabeled data in the training set was minimized, and redundant targets were factored out.

On the polarized input data, the dummy classifier using the strategy “most frequent” achieved a prediction accuracy of 88 %. The best random forest model trained on the polarized input data achieved a prediction accuracy of 98 %, which was higher than both the models trained on the original data and on the data transformed into a binary trait. When fitting the model to the test data, it correctly predicted all 46 susceptible accessions and misclassified only one of the six resistant accessions.

3.3.3 Identifying consistently high-ranked features

To test the hypothesis whether it was possible to find reliable associations between the k -mers and the phenotype, the feature importances provided by the RF model trained on the polarized input data were investigated. Of the 100,000 k -mers, 647 had a feature importance above zero (0.65 %). The k -mer AAATTAATATGAACCTTGGTATTAGACTTAA had the highest feature importance of 0.46 %. When sorted by their feature importance, the first ten k -mers had a cumulative feature importance of 3.2 %. The first 50 k -mers had a cumulative feature importance of 13.8 %. None of the k -mers with a non-zero gain were also found by the k -mer-based GWAS.

An important assumption for being able to interpret the feature importances was that the k -mers and their assigned feature gain measurements reported by the RF models were being consistent. In a random forest, trees are grown by random selection of features at each node (Breiman, 2001). For this reason, models trained with the same hyperparameters can differ regarding their performance, internal parameters, and generated metrics. To ensure reproducibility, the random state is set to a defined seed before training a model.

To assess the comparability of feature importance measurements from random forest models across multiple runs, 100 models were trained on the polarized input data. Each model used identical hyperparameters but a different seed for the random state.

After each model run, the set of k -mers exhibiting non-zero feature importance was compared against those identified in previous runs. Examination of the intersecting k -mers revealed minimal overlap among the features selected by successive models. For instance, the first model identified 624 k -mers with non-zero feature gain, of which only 34 were also selected by the subsequent model. In the following comparison, only one k -mer overlapped between runs, and this single feature was absent in the next intersection. Overall, the union of the non-zero feature-importance k -mer sets across 100 runs contained 15,339 distinct k -mers, corresponding to 15.34 % of all k -mers in the input dataset. Figure 16 illustrates the number of overlapping and

distinct k -mers as additional random forest models were trained. Because of the large number of k -mers obtained over all runs and the rapid drop in k -mers consistently relevant in the models, it was not possible to robustly identify k -mers or genes using the machine learning method.

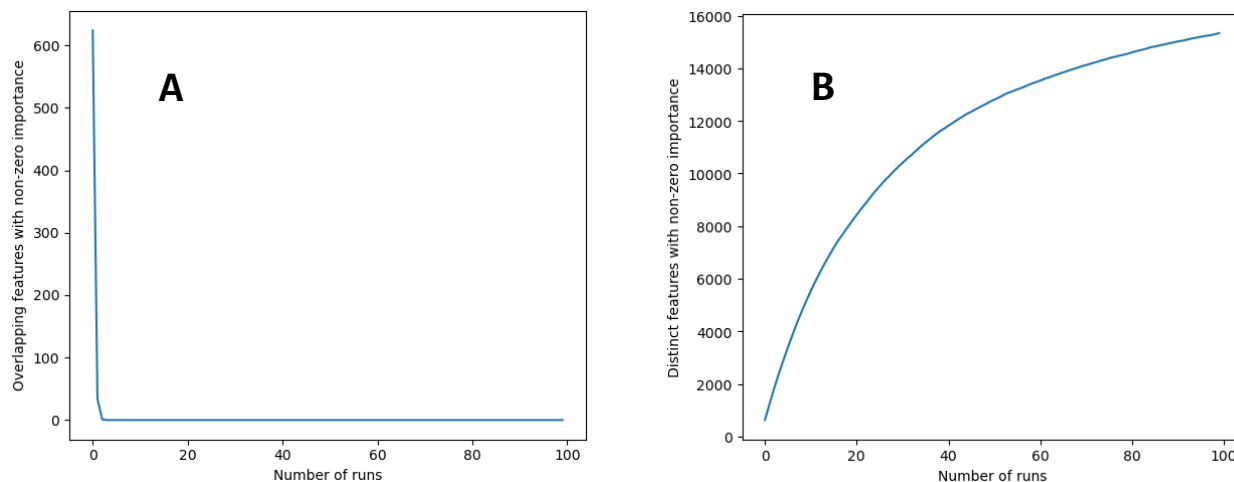


Figure 16: Feature importance consistency assessments. Plots are displaying (A) the number of overlapping k -mers obtained by taking the intersection of the k -mer sets, and (B) the number of distinct k -mers obtained by taking the union of the k -mer sets, both plotted against the number of RF model training runs.

3.3.4 Dimensionality reduction of k -mer data

To further explore the high-dimensional k -mer data, a principal component analysis (PCA) was performed. The first principal component (PC) explained 6.71 % of the variance, while the second PC explained 2.63 %. When plotting this two-dimensional data and labeling the data points according to their resistance phenotype, clusters of accessions with higher k -mer similarity were revealed (Figure 17). The PCA indicated that accessions labeled as “highly susceptible” and “susceptible” clustered together. Accessions in the “medium resistance” group were also closer to the susceptible ones, showing some overlap. In contrast, “high resistance” accessions formed a separate cluster.

The PCA results confirmed the assumption that a sharper separation between susceptible and resistant accessions could be achieved by excluding accessions with labels in the middle of the disease resistance spectrum. Additionally, it supported the observation that there was an overlap between accessions classified as “highly susceptible” and “susceptible”, suggesting that removing ambiguously labeled accessions would further polarize the data.

Another dimensionality reduction technique, linear discriminant analysis (LDA), produced results similar to those of the PCA. When plotting the LDA results in a lower-dimensional space,

the accessions classified as "highly susceptible," "susceptible," and "medium resistance" overlapped once again (Appendix B, Figure 18). Accessions classified as "high resistance" again formed a distinct cluster. This further strengthened the notion that accessions from the first three classes were difficult, if not impossible, to separate using the available *k*-mer data. Additionally, both PCA and LDA results were also plotted in a three-dimensional space, yielding the same observations (Appendix B, Figure 19).

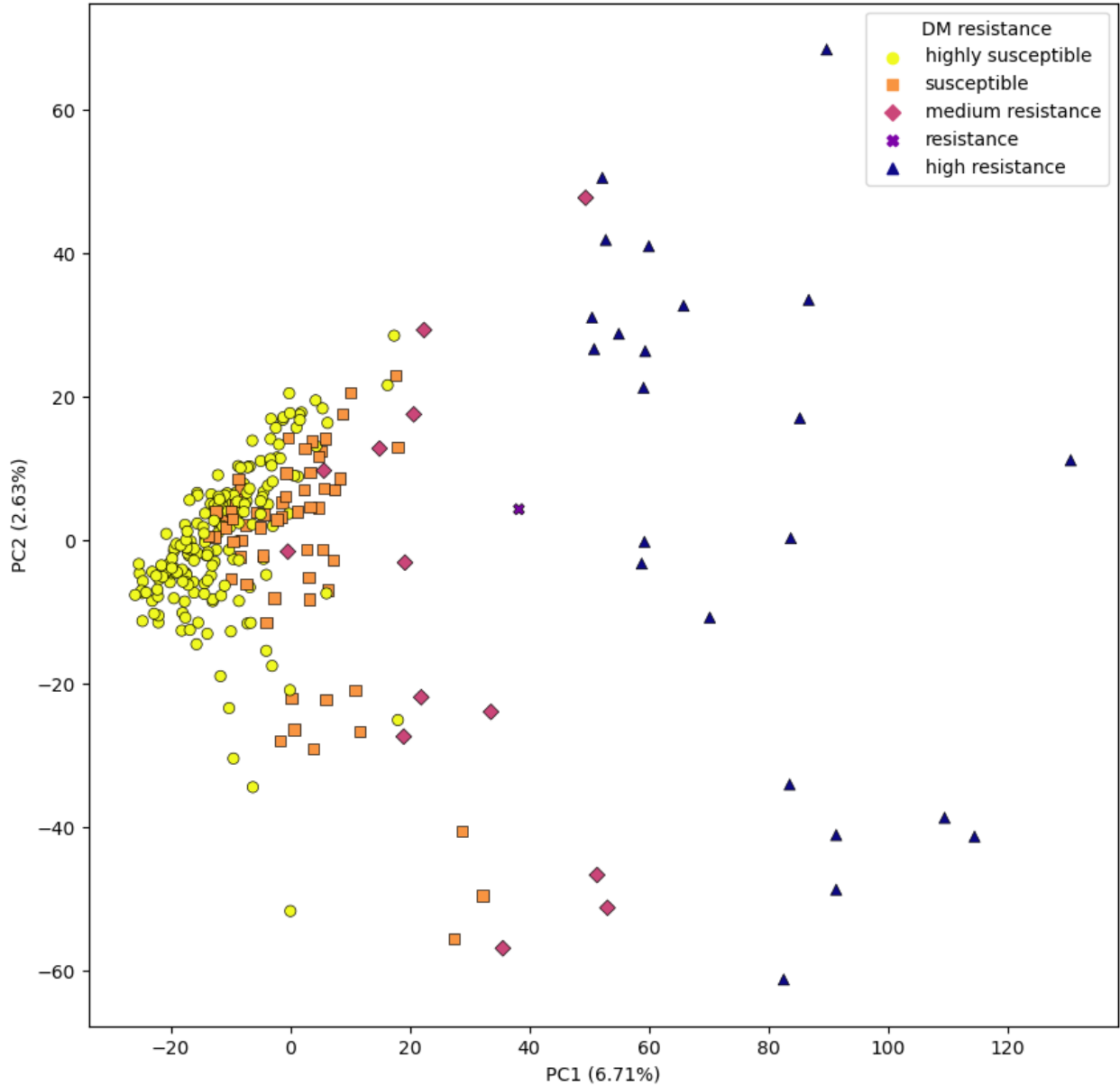


Figure 17: Principal component analysis based on presence/absence information of the pre-filtered *k*-mer data from 295 accessions, related to five downy mildew resistance classes. The first two principal components (PC1: 6.71 % variance, PC2: 2.63 % variance) separate samples into different resistance classes. "Highly susceptible" (yellow circles) and "susceptible" (orange squares) samples cluster together, while "medium resistance" (red diamonds), "resistance" (purple asterisks), and "high resistance" (blue triangles) samples are more distinct, with "high resistance" samples showing the greatest dispersion along PC1.

4 Discussion

In the past decade, genome-wide association studies have established themselves as a generally powerful and reliable tool for researching complex genotype-phenotype associations in plants. However, the full potential of GWAS has been impeded by several challenges, including identification of rare variants, spurious associations and effectively controlling for false positives, and computational efficiency (Tibbs Cortes et al., 2021) in the light of steadily increasing sample sizes (Abdellaoui et al., 2023).

In recent years, k -mer-based methods have been utilized for a wide range of purposes in the field of bioinformatics (Moeckel et al., 2024). Among them, k -mer-based association studies have emerged as alternative methods, successfully addressing some of the limitations of traditional SNP-based GWAS (M.-M. Chen et al., 2023; Kim et al., 2020; Lemay et al., 2023; Voichek & Weigel, 2020). Notable advantages include the ability to find larger structural variants (which could be absent in the reference genome and thus cannot be called but can be captured by k -mers) and that they allow for reference-free association tests. The present study aimed at evaluating the effectiveness of a k -mer-based GWAS, comparing it with traditional association tests, and to complement it with machine learning methods.

4.1 Optimizing k -mer parameters

Several parameters of the k -mer-based GWAS required careful consideration. In this study, the length $k = 31$ was chosen for all k -mers analyzed. This is a commonly chosen value which has been demonstrated to yield good results in previous studies (Lemay et al., 2023). Odd k -mer lengths help avoid ambiguous sequences, as they have distinct canonical and non-canonical forms, unlike sequences with even lengths where both forms can be identical. Among odd values, 31 is the largest k -mer length that still fits in a single 64-bit CPU register. This offers higher sensitivity (due to more possible permutations) while maintaining computational efficiency (Díaz-Domínguez et al., 2024).

Very long k -mers could also lower the detection rate of the GWA. Multiple closely spaced variants can become captured within the same k -mer, effectively tying their signals together and diluting truly causal variants. The challenge is to find k -mers short enough to provide a fine resolution while having sufficient mapping specificity (due to many identical short k -mers scattered around the genome). However, previous studies have also used 25-mers in k -mer-

based GWAS (Voichkek & Weigel, 2020). Repeating the analyses with different values of k would be necessary to make definite statements about the practical impact of the k -mer length.

4.2 Interpreting the performance and utility of k -mer-based GWAS

The results indicate that while k -mer counting represents the most computationally demanding step unique to k -mer-based GWAS, the overall association testing process is significantly faster than traditional SNV-based GWAS. The increased runtime for downstream k -mer analysis was primarily due to searching large raw sequencing datasets for significant k -mers. However, the development of a specialized tool within this study effectively mitigated this bottleneck, improving efficiency compared to naïve search methods.

In SNV-based GWAS workflows, genotype calling typically constitutes the computational bottleneck due to its complexity and resource demands (T. Wang et al., 2024). As the variants used in this study originated from Nguyen-Hoang et al. (2024), exact runtime measurements of the variant calling step are unavailable. Nevertheless, variant calling for datasets of comparable size can be estimated to take multiple days when running parallelized on a similarly performant computing cluster. Execution of the genome-wide association tests using k -mers took only around a quarter of the time compared to tests based on SNVs, despite the added computational load introduced by permutation testing. Although the association testing step in k -mer-based GWAS was faster than SNV-based GWAS, downstream analysis introduced computational challenges.

In conclusion, the k -mer-based approach offers the advantage of rapidly obtaining results without relying on a reference genome. However, interpreting and localizing significant k -mers is more complex compared to SNVs, which are already annotated with positional information. To infer the effect of k -mer variants on a biological level, their precise genomic locations must first be established.

Despite these challenges, significant k -mers can still function effectively as molecular markers without requiring mapping to a reference genome. Therefore, they can be utilized for marker-assisted selection to efficiently screen individuals carrying desired traits within populations. Previous studies have demonstrated the success of this cost-effective approach in improving pathogen resistance, abiotic stress tolerance, and nutritional quality (Thudi et al., 2021).

Future work could focus on refining k -mer-based methods for more efficient effect annotation and integration with existing variant-based approaches. Additionally, tailoring existing and

new algorithms specifically for k -mers could further enhance their applicability to large-scale breeding programs and improve the overall usability of k -mer GWAS, for instance by reducing computation times.

4.3 Implications of findings on downy mildew resistance

The k -mer-based GWAS was able to identify significantly associated k -mers for 16 out of 20 traits, demonstrating its utility across multiple phenotypes. Downy mildew, caused by oomycetes in the *Peronosporaceae* family, poses a major threat to spinach production worldwide, with new pathogen races emerging in recent decades (J. C. Correll et al., 2011). This disease poses a major threat to spinach production worldwide and has a high negative impact on product quality (Klosterman, 2016). Because of its economic importance and potential for genetic control, DM resistance trait was the primary focus of this study.

The k -mer-based GWAS had well-supported associations on chromosome 3 for the nucleotide-binding site leucine-rich repeat (NBS-LRR) resistance gene *SOV3g001250* and two genes surrounding it, *SOV3g000990* (annotated as coding for a disease resistance protein) and *SOV3g001240* (unknown protein). This replicated the findings of the SNP-based GWAS in the study by Cai et al. (2021), which found a strong signal around the same region at the start of chromosome 3. Notably, they identified the resistance gene *SOV3g001250* to be a potential candidate for conferring downy mildew resistance. Both genes *SOV3g001250* and *SOV3g000990* were reported by Cai et al. to be R genes in the Monoe-Viroflay genome. This region, conferring resistance to *P. effusa*, is known as *RPF3* locus (Bhattarai et al., 2022).

The SNV-based GWAS conducted in this study complements the earlier findings, revealing a long stretch of significant variants centered around the aforementioned disease resistance genes. Investigating this larger region at the start of chromosome 3, additional interesting genes are revealed, including six (putative) disease resistance proteins. Close to a major disease protein is *SOV3g001280*, encoding a cysteine and histidine-rich domain-containing protein RAR1. This signaling gene has been identified to be an essential component in the signaling cascade required for activation of NBS-LRR resistance proteins (Muskett et al., 2002). The nearby gene *SOV3g000770* encodes a NAC domain-containing protein. NAC transcription factors are involved in both abiotic stress responses and pathogen resistances (Han et al., 2023). The results of this study highlight the multiplicity of the *RPF3* downy mildew resistance locus, suggesting that it includes not only resistance genes but also genes potentially involved in regulatory or structural functions related to resistance.

Two genes, *SOV3g000280* and *SOV3g000380*, express cellulose synthases that influence cell wall composition. Both *SOV3g0001380* (a 3-ketoacyl-CoA synthase) and *SOV3g00040* (a PKS_ER domain-containing protein) participate in wax biosynthesis, potentially affecting cuticle permeability and thus enhancing abiotic stress tolerance and protect against pathogens (Luo et al., 2024; Sun et al., 2023). Overall, these results emphasize the bipartite nature of disease resistance in spinach, consisting of a synergy of physical and biochemical mechanisms.

Cai et al. (2021) also point out a strongly associated SNP located within a gene on chromosome 4. This gene is coding for a wax ester synthase, having them theorize about the possible implications regarding their role in reinforcing the physical barrier of the plant and protection from pathogens. Their results align with the current study's *k*-mer-based GWAS, which also identified an enzyme catalyzing wax ester synthesis. Specifically, the gene *SOV2g037000* (annotated as Putative O-acyltransferase WSD1-like), which previously has not been associated to DM resistance in spinach, had strong evidence through *k*-mers. The WSD1 enzyme mainly acts as a wax synthase and catalyzes the biosynthesis of cuticular waxes in plant stems and leaves. The cuticle is a protective, hydrophobic layer covering the epidermis of plants (F. Li et al., 2008). Cuticular waxes have been found to protect from abiotic stress and are considered an important factor in fungal infections and development (Lewandowska et al., 2020). They affect numerous fungal pathogens causing mildew diseases (X. Wang et al., 2020). A gene belonging to the O-acyltransferase WSD1-like family has previously been associated with resistance against a basidiomycete pathogen in barley (Abo-Elyousr et al., 2022). These findings support a possible link between downy mildew resistance and wax ester synthesis in spinach, warranting further investigation.

Moreover, these findings suggest that, alongside disease resistance mechanisms involving proteins belonging to the NBS-LRR family which recognize virulence proteins and activate an immune response (DeYoung & Innes, 2006), genes affecting the physical barrier of the plant might play a role in disease resistance. Especially in the light of newly emerging downy mildew races overcoming the NBS-LRR-based resistances in established spinach cultivars (Bhattarai et al., 2021), it might be worthwhile to pursue alternative resistance strategies. Further research is needed to evaluate their potential in breeding programs.

Heatmaps visualizing the presence/absence patterns of *k*-mers significantly associated with a particular trait can assist in identifying potential markers for marker-assisted selection, allowing for efficient screening of individuals carrying the desired traits within breeding populations. Notably, this approach has been applied to improving disease resistance, but also abiotic stress responses and nutritional qualities (Thudi et al., 2021).

4.4 Considerations and limitations of GWAS approaches

Although the *k*-mer-based GWAS replicated findings from SNV-based studies and identified a promising candidate gene, the results must be interpreted with caution due to the potential for false positives, limited resolution in genomic localization, and the challenges of biological interpretation. As previously mentioned, genome-wide association studies carry several limitations that even the *k*-mer-based approach cannot fully address. Notably, while association testing identifying significant *k*-mers is indeed a reference-free method, the step in the downstream analysis of mapping extracted reads does require a reference genome.

4.4.1 Rare variant detection

A restriction applying to both traditional and novel GWAS is the value of the minor allele frequency/minor allele count. A balanced trade-off must be found between a too lenient filtering which can include sequencing artifacts and reduce statistical power, and a too strict criterium that possibly excludes rare variants.

Although the sample size used in the GWAS was quite big, the number of samples in some classes of categorical traits was below 16. Since the minor allele count threshold was set to 16, variants unique to accessions in small classes would be filtered out. However, three of those classes (“erect” plant type, “halberd” leaf shape, and “resistance” DM resistance) contained only a single accession, which is insufficient for definitive association analyses and thus cannot be reliably analyzed due to insufficient sample size. The “medium resistance” class of the DM resistance trait contains 13 accessions, which was just below the threshold. However, because the ordinal variable describing the resistance status was treated as continuous by the linear mixed model, it still influences the association tests together with the “resistance” and “high resistance” classes. Furthermore, *k*-mers found for majority classes could have indirectly revealed associations of rare phenotypes (e.g., significant *k*-mers identified in majority classes could highlight genomic regions where minority-class accessions carry alternative variants, indirectly suggesting associations with rare phenotypes).

4.4.2 Sequencing and data quality considerations

In a GWAS, it is important to ensure that the sequencing coverage is high enough to capture nearly all existing variants. Specifically in the case of a *k*-mer-based GWAS, the *k*-mers counted for a sample should be close to “saturation”, i.e., increasing the coverage should not lead to

strikingly increased k -mer counts. Otherwise, k -mers erroneously labelled as missing in some accessions could negatively affect the association outcomes.

When plotting the total number of counted k -mers against sequencing coverage per accession, accessions US355 (SRR15321797) and US356 (SRR15321665) appeared as outliers, showing much higher k -mer counts than all other samples. The relatedness matrix also indicated a close relationship between these two accessions. One potential explanation is contamination of the sequencing material, resulting in additional k -mers counted from sequencing reads stemming from foreign genetic sources. Since both accessions represent *S. tetrandra* from Georgia, Europe, and were presumably sequenced simultaneously, batch effects during sequencing could also explain the observed discrepancy. The two accessions were not excluded from the data, because they only had phenotypic data for the flowering and bolting time traits and rare k -mers got filtered by the minor allele frequency threshold.

It is a commonly employed practice to trim raw sequencing reads to remove adapter sequences and low-quality bases, although the necessity of this step is under dispute (S. J. Bush, 2020). Read trimming was not performed for the k -mer-based GWAS. Although this approach may introduce a small risk of random errors affecting variant calls at the population level, it should not have caused issues during the downstream read mapping stage, because Bowtie 2 performs local alignments and clips characters from read ends to maximize alignment scores. Besides that, rare sequencing errors were likely filtered by the MAF threshold.

4.4.3 K -mer downstream analysis and read mapping interpretations

Although k -mer-based GWAS delivered promising results, the absence of specialized downstream analytical tools remains a limitation, particularly for interpreting significant k -mers. Processing large amounts of raw sequencing data—with datasets reaching up to multiple terabytes in size—creates a substantial computational burden, especially when extracting reads containing these significant k -mers. The specialized tool developed in this study is specifically optimized to handle k -mers and generate useful match statistics.

A notably high proportion of sequencing reads mapped to multiple genomic positions (~80 % for DM resistance). Differentiating between spurious and genuinely causative mappings seriously complicates result interpretation. To mitigate this issue, the number of mapped reads was employed as a proxy for importance of a region in the genome. When visualizing these findings through novel k -mer-based Manhattan plots, data points were plotted with varying transparency levels, highlighting genomic regions with higher reads support. Determining the

optimal transparency value, however, remains challenging and can vary substantially across different trait-specific GWAS analyses. Setting the transparency too low can supposedly amplify noise, while an overly strict threshold hides potentially relevant results.

To identify hypothetically important genes, annotated genomic regions were sorted by the number of mapped reads extracted. Although straightforward, this method effectively distinguishes genes with substantial supporting evidence from those with limited evidence. Nevertheless, this sorting strategy remains approximate and does not strictly quantitatively express significance, as it can be heavily influenced by variability in sequencing depth across and within accessions. Higher sequencing depth in certain regions naturally results in an increased count of mapped reads, potentially skewing interpretations. Normalization of coverage could address this issue, although it potentially leads to a loss of information. Conversely, if a significant k -mer was rare and present in only a small percentage of samples, there would have been fewer reads extracted and mapped to an annotated region; these smaller signals might become obscured by reads extracted from more commonly occurring k -mers. Nevertheless, looking at the results of the GWAS downstream analysis, there was usually a marked gap between some genes with very high mapping counts and a longer list of genes with only a few reads mapping to them.

Another considerable challenge arises from the fact that roughly 70 % of DM resistance-related reads are mapped to regions lacking gene annotations. Non-coding DNA can have important regulatory functions, for example altering gene expression through modulated transcription factor binding or post-transcriptional effects. These effects can impact phenotypes (Peña-Martínez & Rodríguez-Martínez, 2024). Although these underlying k -mers might indicate biologically important regions such as transcriptional regulators or unidentified genes, assessing their biological function remains difficult with currently available tools. Visualization through k -mer-based Manhattan plots helps to highlight prominent genomic regions or chromosomes of interest. This was demonstrated by showing that a region at the start of chromosome 3 had very strong signals and is expected to play a crucial role in downy mildew resistance.

Although the fraction of unmapped reads for DM resistance was small (~4 %), these represent another analytical challenge. One plausible explanation for their failure to align could be bigger differences from the reference genome, perhaps due to large structural variations such as insertions or deletions. Identifying such variants is particularly important, as they are often missed by traditional SNV-based GWAS methods but can substantially influence phenotypic traits (Saxena et al., 2014).

The spinach reference genome was sequenced from the downy mildew-susceptible inbred cultivar Monoe-Viroflay (Bhattarai et al., 2022). Consequently, reads derived from resistant accessions may fail to align due to structural variations, such as large insertions or deletions absent from the reference. Previous studies have shown that assembling overlapping k -mers into longer sequences prior to mapping can capture structural variations (Voichek & Weigel, 2020), and it might also improve sensitivity when extracting the sequencing reads. Although preliminary attempts were made in this study to assemble reads from overlapping k -mer sequences, these tests were unsuccessful and were not pursued further.

4.4.4 Statistical modelling limitations using discrete traits

For the SNV-based GWAS, **GEMMA** was selected to perform association tests, because it was already part of the k -mer-based pipeline, is commonly employed in GWAS analyses, and allows the incorporation of a relatedness matrix to adjust for population structure. Correcting for population structure using linear mixed models is critical, as it addresses kinship as a confounding factor (Hoffman, 2013).

A limitation affecting both approaches is that the LMMs used by **GEMMA** assume a linear relationship between predictor and response, constant variance of residuals (homoscedasticity), and normally distributed residuals (H. Chen et al., 2016; Schielzeth et al., 2020). For ordinal or binary traits, phenotype data is non-continuous, and the relationship between predictor and response typically will not be linear. Additionally, assumptions regarding the distribution of the residuals are likely not met. Consequently, applying LMMs to discrete traits can lead to an increase of the type I error rate, i.e., the occurrence of false positives (W. Zhou et al., 2018), and reduced statistical power.

One way to address this is to manually transform the phenotype data into continuous traits to better represent the actual biological relationship between the classes (Yang et al., 2023). However, this approach is problematic because the underlying relationship is usually unknown and thus cannot be accurately represented. Furthermore, estimating a latent continuous phenotype from discrete data using the same genetic variants subsequently tested for association introduces circular analysis, potentially causing bias. Another solution is to switch to methods which explicitly model nonlinear effects, like generalized linear mixed models. These models are better suited for discrete traits, avoiding an inflated rate of false positives (H. Chen et al., 2016).

Nevertheless, LMMs have demonstrated robustness and fit a large variety of datasets, even to ones that deviate slightly from the standard (Schielzeth et al., 2020). Although the consequences of violating model assumptions stay generally within reasonable bounds, it can lead to misleading results in some cases and must be kept in mind when interpreting GWAS results.

4.4.5 Strategies for handling multiple testing in GWAS

Due to the vast number of concurrent statistical tests performed in a GWAS, the risk of falsely rejecting a true hypothesis (committing a type I error) rapidly becomes very high when settling on conventional alpha levels such as 0.05, 0.01, or even 0.001. To control for false discoveries at the genome-wide level, appropriate significance thresholds for the family-wise error rate must be established. Among various methods, the Bonferroni correction is the oldest and most commonly used procedure. Although criticized for being overly conservative, Bonferroni correction is favored for its simplicity (Goeman & Solari, 2014).

The Holm method, a sequential adaptation of the Bonferroni correction, is strictly more powerful, allowing the rejection of at least as many hypotheses as Bonferroni, if not more. Despite Holm's theoretical superiority, the simplicity and convenience of the Bonferroni method often remain deciding factors (Goeman & Solari, 2014). Furthermore, Holm's method is not directly integrated into GEMMA and does not define a single universal significance threshold, complicating the visualization through Manhattan plots. Consequently, the Bonferroni correction was selected for the SNV-based GWAS, using an FWER level of 0.05. Given the dataset size, a more stringent level such as 0.01 might have also been appropriate, but it would carry the risk of overlooking potentially important genetic variants.

In the case of k -mer-based GWAS, a challenge is clearly defining the set of hypotheses over which the multiple testing correction needs to be performed—a recurring problem in multiple testing (Bender & Lange, 2001). This issue arises because a single variant can be represented by up to k distinct k -mers, resulting in high correlation among the tested k -mers. The exact number of independent tests is unknown, making corrections like Bonferroni and related methods overly conservative. To address this, permutation testing was employed. By repeatedly shuffling phenotype labels and recalculating test statistics, permutation testing empirically establishes significance thresholds that implicitly reflect characteristics of the dataset and consider the correlation structure inherent in the dataset (Churchill & Doerge, 1994). In this case, permutation testing leads to a less strict but statistically more powerful approach than Bonferroni correction. Permutation testing is considered the gold standard in multiple testing

correction in GWAS but suffers from being computationally highly demanding (Tibbs Cortes et al., 2021).

A parallel problem can be found in genetic association studies based on SNPs and indels. Genetic variants located physically close to each other on a chromosome tend to be inherited together more frequently than expected by chance, a phenomenon known as linkage disequilibrium (LD). LD results in high correlation among variants, thereby decreasing the number of independent tests. Despite this correlation, SNP-based GWAS studies have historically proven effective at identifying genuine genotype-phenotype associations (W. S. Bush & Moore, 2012).

Although the redundant nature of counted k -mers inflates correlations in the data, it also provides an advantage: it increases the probability that at least one of the multiple possible k -mers representing a true causal variant is distinct enough across the genome to show a clear association with the phenotype. Therefore, it increases the sensitivity of identifying biologically relevant variants.

4.5 Integration of machine learning for phenotype prediction

Dimensionality reduction techniques such as principal component analysis (PCA) and linear discriminant analysis (LDA) highlighted the phenotype structure in the data and suggested the presence of groups of accessions likely separable by advanced machine learning algorithms. While PCA is an unsupervised method useful for rapidly exploring data structure, LDA uses class labels to maximize the distance between groups. Although PCA has known limitations when applied to genetic data, it is an established method with a long track record of accurately describing population structure (Abegaz et al., 2019).

RF models are widely used in bioinformatics due to their effectiveness in handling high-dimensional data and their robustness against bias. Furthermore, RF models provide additional measurements, such as feature importance values, which can be used to assess the informative value of the input data and rank features accordingly (Salman et al., 2024). This study demonstrates that random forest machine learning models can be effectively trained on k -mer presence/absence counts to predict the resistance status accessions, potentially serving as molecular markers for specific phenotypes. Model performance was dependent on data preprocessing; specifically, models that were trained exclusively with polarized input data (transformed to a binary trait) achieved higher prediction accuracies compared to models trained on ambiguously classified samples. Even with highly imbalanced datasets, in which case

the majority class outnumbered minority classes nearly eightfold, excellent prediction accuracies were attainable. To avoid a misleading representation of the results, it is important to use a dummy classifier as a benchmark, which properly reflects the class imbalance.

Due to the random nature of the RF models, extraction of k -mers with the highest feature importances assigned by the model was not feasible. Variants with above-zero feature importances were inconsistent and showed little overlap between model training runs. While this limits the ability of using random forest ML models to pinpointing variants strongly linked to traits, they nevertheless predicted the disease resistance status of samples with high accuracy.

An approach to address this issue could be hyperparameter optimization using automated tuning such as implemented in **Hyperopt** (Bergstra et al., 2013). Instead of evaluating combinations of parameters in a defined grid, **Hyperopt** employs Bayesian optimization to search more efficiently over much larger search spaces. This way, parameters concerning the maximum number of features per split or the number of trees could increase the total number of features considered by the RF model, potentially enhancing marker identification. Coupled with averaging feature importances or only retaining features with the highest scores across numerous independent model training runs, this strategy could enhance the robustness and reproducibility of the identified molecular markers.

5 Summary and Conclusions

Spinach (*Spinacia oleracea*) is a leafy green vegetable that is cultivated worldwide for its nutritional qualities and health benefits. Spinach production has a high economic importance but is persistently threatened from downy mildew (DM), a destructive disease caused by *Peronospora effusa*. Genetic resistance remains the most effective and sustainable strategy for disease management; however, rapidly evolving pathogen races continuously overcome existing resistances, necessitating the identification of novel resistance mechanisms and genetic markers.

This thesis addressed this challenge by performing genome-wide association studies (GWAS) on a diverse panel of 303 spinach accessions, comprising both cultivated varieties and wild relatives. Two GWAS methodologies were applied and compared: a traditional SNV-based GWAS and an innovative *k*-mer-based approach, which is independent from a reference genome. The primary objective was to identify genomic regions associated with 20 key agronomic traits, focusing primarily downy mildew resistance.

Based on the results of this study, several conclusions can be drawn:

1. Known DM resistance genes in spinach have been validated—Both conventional and *k*-mer-based GWAS approaches have confirmed significant associations with previously identified resistance genes located at the start of chromosome 3. This locus contains several disease resistance proteins, including *SOV3g001250* and *SOV3g000990*, corroborating earlier GWAS findings.
2. Discovery of a novel resistance-associated gene—The *k*-mer-based GWAS identified *SOV2g037000*, a gene on chromosome 2 which has not yet been linked to DM resistance. This gene encodes a putative O-acyltransferase WSD1-like protein, which is involved in cuticular wax synthesis and may represent an alternative resistance mechanism.
3. Prediction of resistance phenotypes using machine learning—Random forest models trained on *k*-mer data classified spinach accessions as resistant or susceptible with 98 % accuracy, demonstrating the potential of machine learning methods to quickly and accurately predict the resistance phenotype without reliance on a reference genome.
4. Distinguish the advantages and limitations of *k*-mer based GWAS—This method successfully identified molecular markers for most agronomic traits and detected potential variants that might be missed by traditional SNV-based approaches. However, practical limitations were identified, particularly in *k*-mer downstream analysis. Mapping significant *k*-mers to precise genomic locations and interpreting results is

hindered by a lack of specialized tools and robust methods. As part of this thesis, a custom program was developed to facilitate *k*-mer downstream processing.

5. Introduction of *k*-mer-based Manhattan plots—This study presents a new visualization strategy for *k*-mer-based GWAS results, displaying *p*-values of significantly associated *k*-mers contained in extracted sequencing reads mapped to locations in the reference genome. These plots function similarly to conventional Manhattan plots, aiding interpretation of the results.

5.1 Implications and future directions

The findings of this thesis highlight the utility of *k*-mer-based GWAS approaches in plant genomics, particularly for species with limited genomic resources or complex variant landscapes. By confirming a known resistance locus and identifying a gene not previously associated with downy mildew resistance, this research provides valuable insights for spinach breeding. Furthermore, the successful application of machine learning to *k*-mer data demonstrates a powerful approach for rapid prediction of resistant genotypes.

Further research is needed to fine-tune *k*-mer-based GWAS methodologies, resolving open concerns mainly related to multiple testing correction and downstream analysis. Additional tools are needed to improve the *k*-mer downstream analysis, and robust methods to enhance interpretation and visualization of results. Alternative disease resistance mechanisms should be in the focus of researchers, especially in the light of evolving pathogens overcoming established resistances in spinach.

In summary, this study emphasizes the effectiveness of *k*-mer-based GWAS and machine learning approaches for discovering genotype-phenotype associations in spinach. The outcomes of this study may assist future resistance breeding efforts in spinach, addressing ongoing agricultural challenges.

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Declaration of the use of generative AI tools

In writing this master's thesis, I used ChatGPT to proofread and enhance the stylistic elements of some formulations in my text. GitHub Copilot was used to assist programming in some of the scripts and parts of the specialized tool. Moreover, I used DeepL to translate an initial version of the German abstract, using the English abstract that I had previously written.

List of abbreviations

Bp	Base pair
DM	Downy mildew
DNA	Deoxyribonucleic acid
FAO	Food and Agriculture Organization
FWER	Family-wise error rate
GWA	Genome-wide association
GWAS	Genome-wide association study
HWE	Hardy-Weinberg equilibrium
IBS	Identity-by-state
IGV	Integrative genomics viewer
Indel	Insertion-deletion
ISF	International Seed Federation
LD	Linkage disequilibrium
LDA	Linear discriminant analysis
LMM	Linear mixed model
MAC	Minor allele count
MAF	Minor allele frequency
ML	Machine learning
NBS-LRR	Nucleotide-binding site leucine-rich repeat
NGS	Next-generation sequencing
PacBio	Pacific Biosciences
PCA	Principal component analysis
QC	Quality control
RF	Random forest
SMRT	Single-molecule real-time
SNP	Single-nucleotide polymorphism
SNV	Single nucleotide variation
SO	Sequence Ontology
SRA	Sequencing Read Archive
UTR	Untranslated region
VCF	Variant Call Format
VSC	Vienna Scientific Cluster

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Appendix A

Table 4: SNPs and indels captured by reads containing *k*-mers significantly associated with downy mildew resistance, mapped to the putative resistance gene *SOV3g001250*. Continues on next page.

Position	Reference base	Alternate base
SOVchr3:1,247,020	T	G
SOVchr3:1,247,022	A	G
SOVchr3:1,247,028	T	C
SOVchr3:1,247,031	T	C
SOVchr3:1,247,040	G	T
SOVchr3:1,247,047	A	G
SOVchr3:1,247,050	A	C
SOVchr3:1,247,052	C	G
SOVchr3:1,247,055	T	C
SOVchr3:1,247,072	T	C
SOVchr3:1,247,073	G	A
SOVchr3:1,247,085	T	A
SOVchr3:1,247,090	A	T
SOVchr3:1,247,097	A	G
SOVchr3:1,247,110	T	A
SOVchr3:1,247,115	C	A
SOVchr3:1,247,122	A	G
SOVchr3:1,247,126	A	G
SOVchr3:1,247,128	G	A
SOVchr3:1,247,130	T	C
SOVchr3:1,247,133	G	T
SOVchr3:1,247,137	T	A
SOVchr3:1,247,139	G	T
SOVchr3:1,247,140	G	C
SOVchr3:1,247,142	T	G
SOVchr3:1,247,145	T	C
SOVchr3:1,247,147	A	G
SOVchr3:1,247,150	C	T
SOVchr3:1,247,153	A	T
SOVchr3:1,247,160	T	G
SOVchr3:1,247,164	T	A

SOVchr3:1,247,166	C	T
SOVchr3:1,247,167	C	A
SOVchr3:1,247,178	G	A
SOVchr3:1,247,179	A	G
SOVchr3:1,247,183	A	T
SOVchr3:1,247,193	A	C
SOVchr3:1,247,199	G	A
SOVchr3:1,247,211	C	T
SOVchr3:1,247,223	T	C
SOVchr3:1,247,232	T	C
SOVchr3:1,247,233	G	A
SOVchr3:1,247,238	C	A
SOVchr3:1,248,994	A	G
SOVchr3:1,248,999	A	G
SOVchr3:1,249,003	A	T
SOVchr3:1,249,004	A	A/T
SOVchr3:1,249,006	A	T
SOVchr3:1,249,024	G	G/A
SOVchr3:1,249,034	T	T/A
SOVchr3:1,249,081	T	G
SOVchr3:1,249,082	C	A
SOVchr3:1,249,082	C	A/G
SOVchr3:1,249,100	A	G
SOVchr3:1,249,104	A	C
SOVchr3:1,249,107	A	T
SOVchr3:1,249,115	G	T
SOVchr3:1,249,133	G	A
SOVchr3:1,249,137	A	Deletion
SOVchr3:1,249,138	G	Deletion
SOVchr3:1,249,139	G	Deletion
SOVchr3:1,249,158	G	G/A
SOVchr3:1,249,186	A	G/A
SOVchr3:1,249,201	G	Deletion
SOVchr3:1,249,202	A	Deletion
SOVchr3:1,249,203	A	Deletion

Appendix B

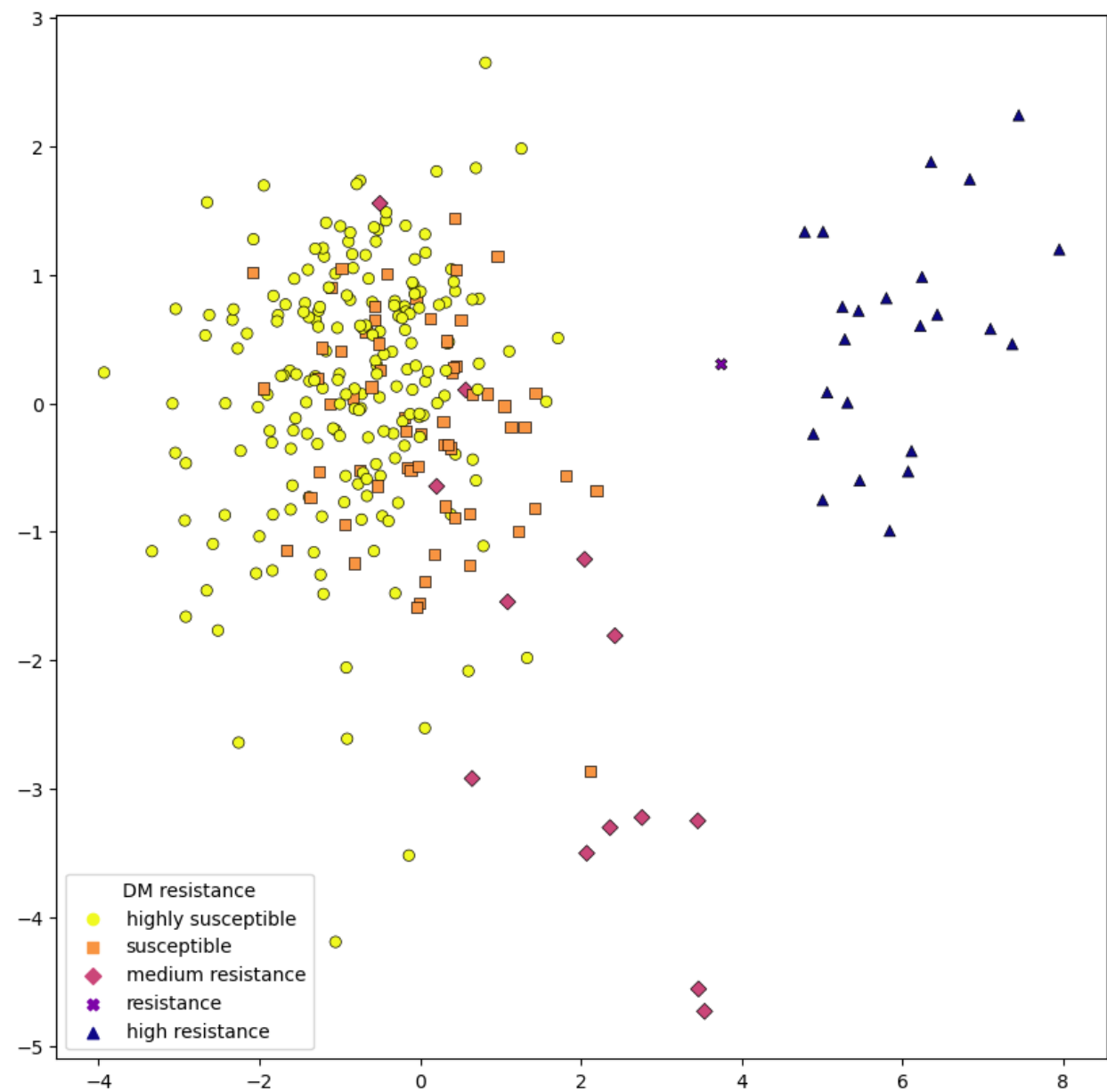


Figure 18: Linear discriminant analysis based on presence/absence information of the pre-filtered *k*-mer data from 295 accessions, related to five downy mildew resistance classes.

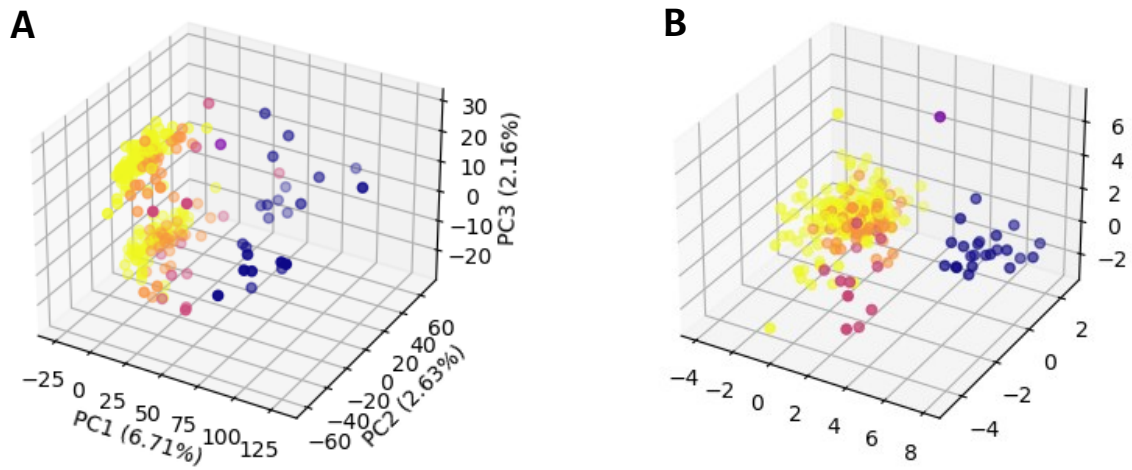


Figure 19: (A) Principal component analysis and (B) linear discriminant analysis based on presence/absence information of the pre-filtered *k*-mer data from 295 accessions. Data points are related to five downy mildew resistance classes, plotted in three-dimensional space.

Appendix C

Table 5: Overview of the computational steps in the k -mer-based workflow. Displayed are the number of threads used for parallelization, the wall-clock runtime in hours, and an estimated or measured single-thread runtime for each process. All listed steps were performed on the Vienna Scientific Cluster.

Process	Threads	Real time (h)	Single-threaded (h)
Sequencing read download	128	39,8	5094,4
Dumping reads to FASTQ format	128	2,9	373,3
K -mer counting	128	38,5	4932,3
Combining and filtering k -mer lists	1	1,6	1,6
Building the k -mer table from lists	1	2,0	2,0
Calculating the k -mer-based kinship matrix	1	1,4	1,4
Running k -mer-based GWA tests for all traits	128	3,2	411,7
Converting significant k -mers to FASTA format	1	< 1,0	< 1,0
Printing sign. k -mer presence/absence patterns	1	< 1,0	< 1,0
Extracting reads with sign. k -mers	128	4,4	567,5
Mapping extracted reads to the reference	48	< 1,0	< 1,0

The SNV-based GWA testing was performed on a high-performance computing cluster with similar hardware, parallelized to 112 threads, taking 17,1 hours wall-clock time or an estimated 1918,0 hours on a single thread.

Appendix D

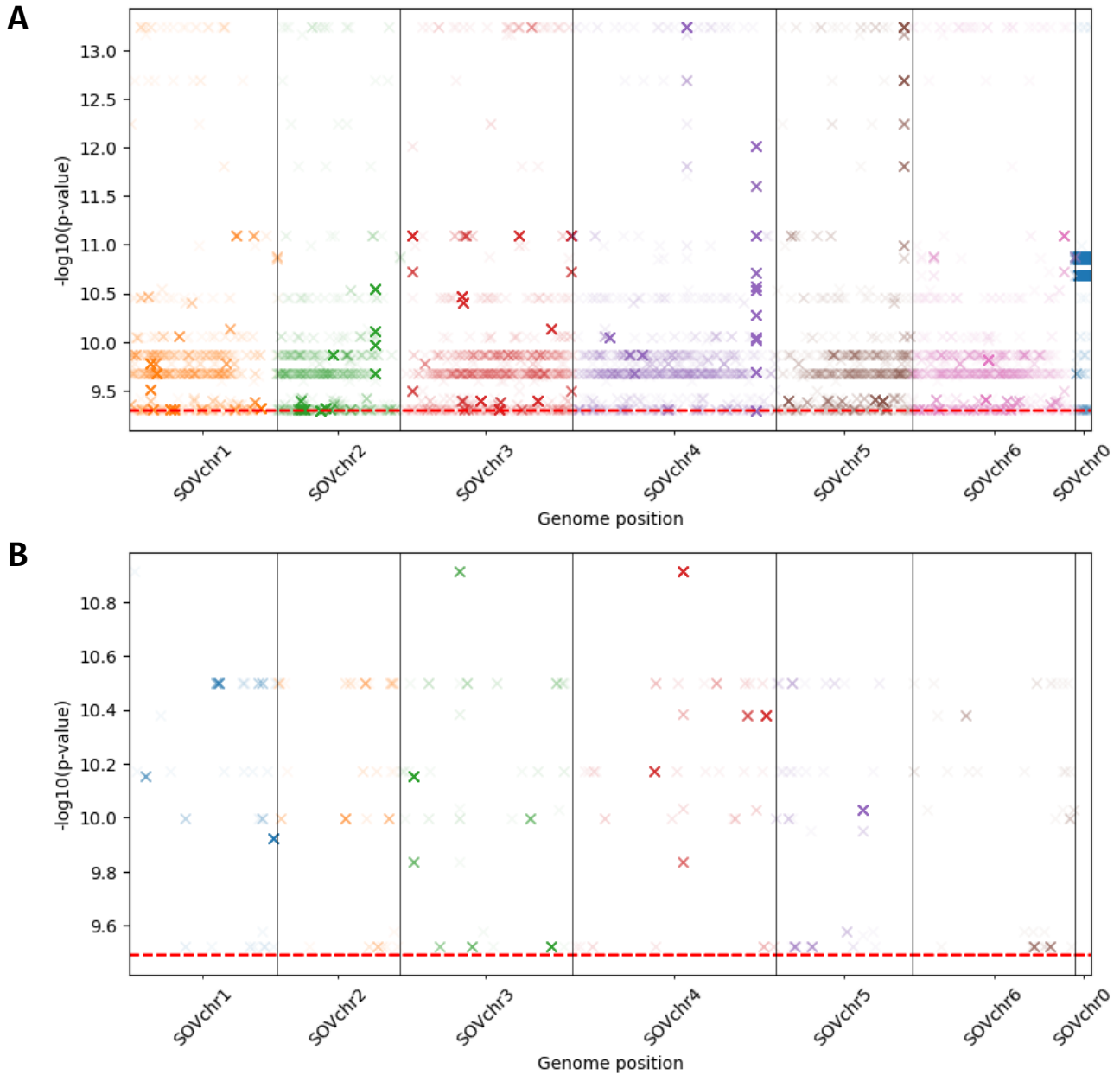


Figure 20: K-mer-based Manhattan plot of GWAS results for (A) plant type and (B) leaf number. Displayed are the lowest p -values of significant k -mers contained in extracted sequencing reads and their corresponding genomic mapping positions, including unanchored scaffolds (collected in “SOVchro”). The dashed red line indicates the genome-wide significance threshold for a 5 % family-wise error rate. The higher the opacity of a data point, the higher the number of supporting mapped reads.

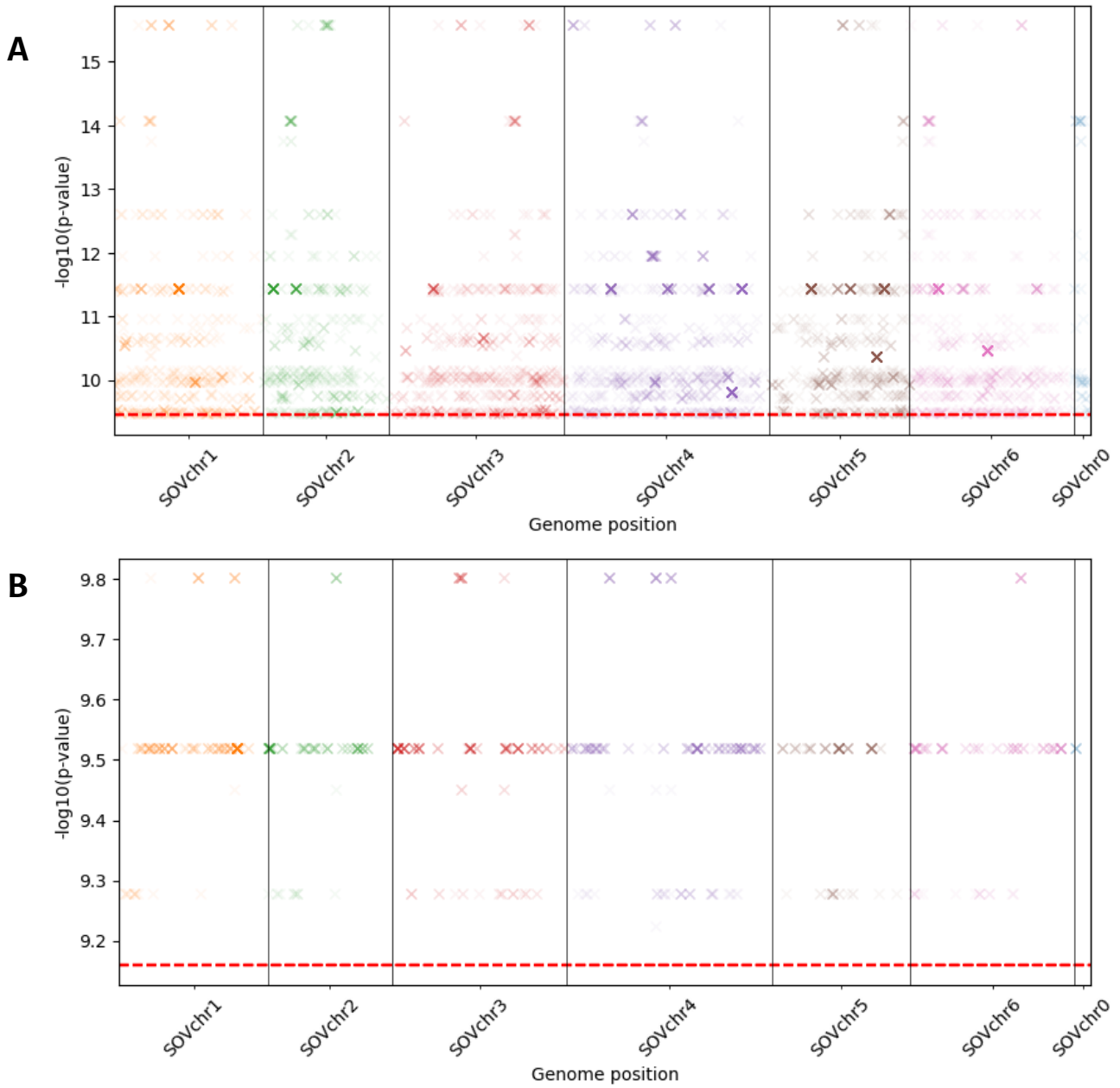


Figure 21: K-mer-based Manhattan plot of GWAS results for (A) petiole width and (B) petiole length. Displayed are the lowest p -values of significant k -mers contained in extracted sequencing reads and their corresponding genomic mapping positions, including unanchored scaffolds (collected in “SOVchro”). The dashed red line indicates the genome-wide significance threshold for a 5 % family-wise error rate. The higher the opacity of a data point, the higher the number of supporting mapped reads.

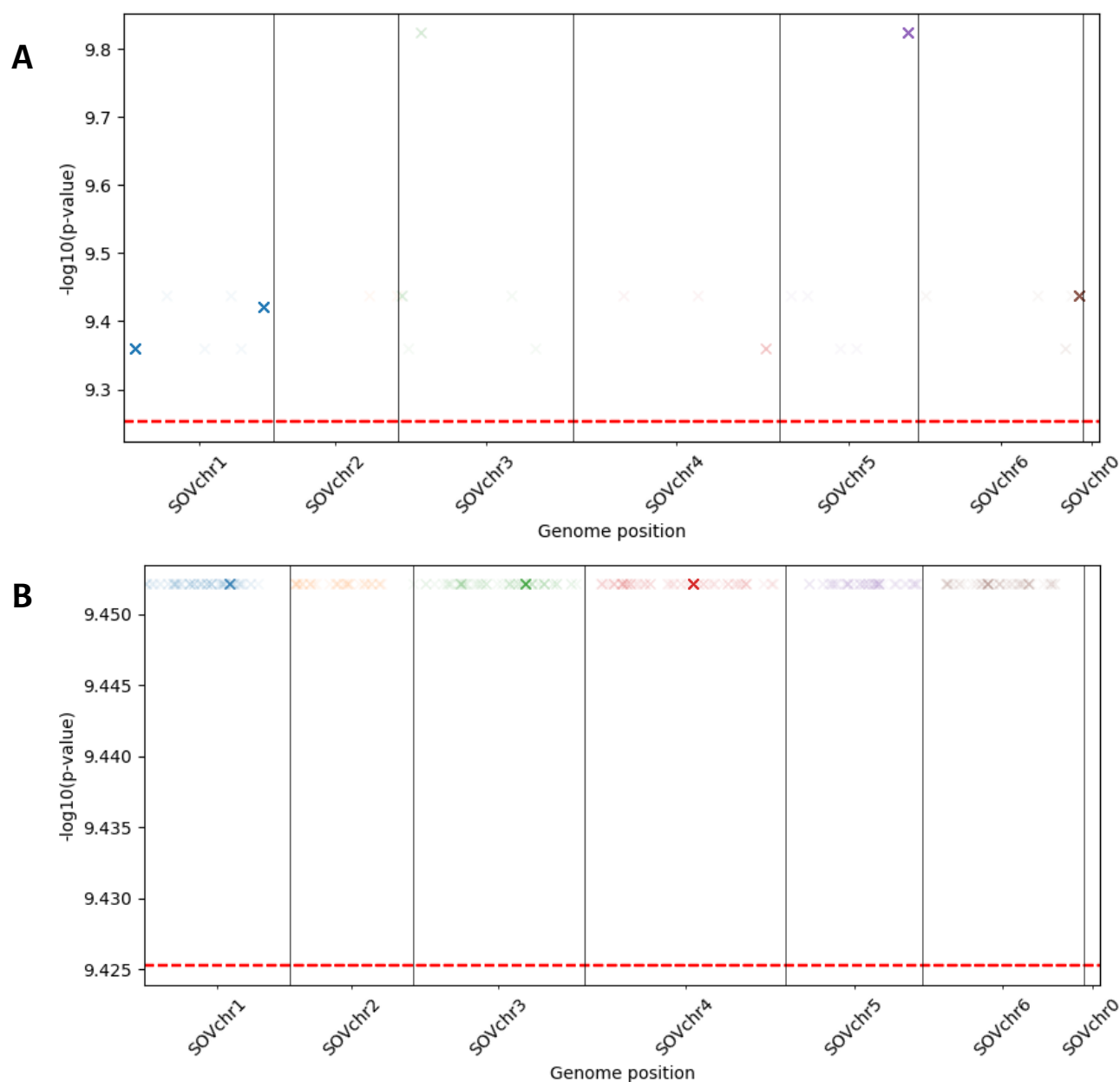


Figure 22: K-mer-based Manhattan plot of GWAS results for (A) petiole color and (B) sex type. Displayed are the lowest p -values of significant k -mers contained in extracted sequencing reads and their corresponding genomic mapping positions, including unanchored scaffolds (collected in “SOVchro”). The dashed red line indicates the genome-wide significance threshold for a 5 % family-wise error rate. The higher the opacity of a data point, the higher the number of supporting mapped reads.

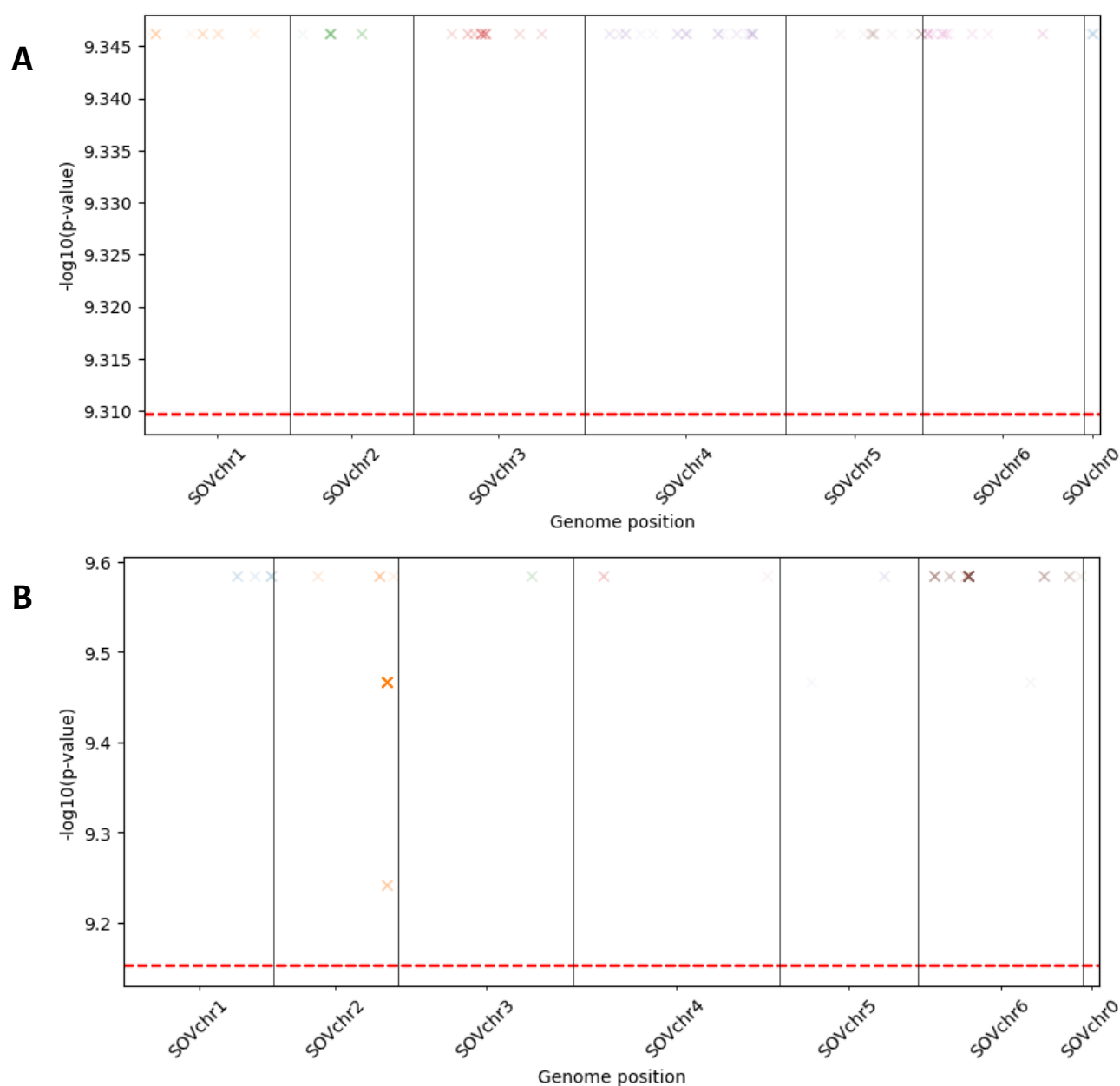


Figure 23: K-mer-based Manhattan plot of GWAS results for (A) leaf shape and (B) leaf surface texture. Displayed are the lowest p -values of significant k -mers contained in extracted sequencing reads and their corresponding genomic mapping positions, including unanchored scaffolds (collected in “SOVchro”). The dashed red line indicates the genome-wide significance threshold for a 5 % family-wise error rate. The higher the opacity of a data point, the higher the number of supporting mapped reads.

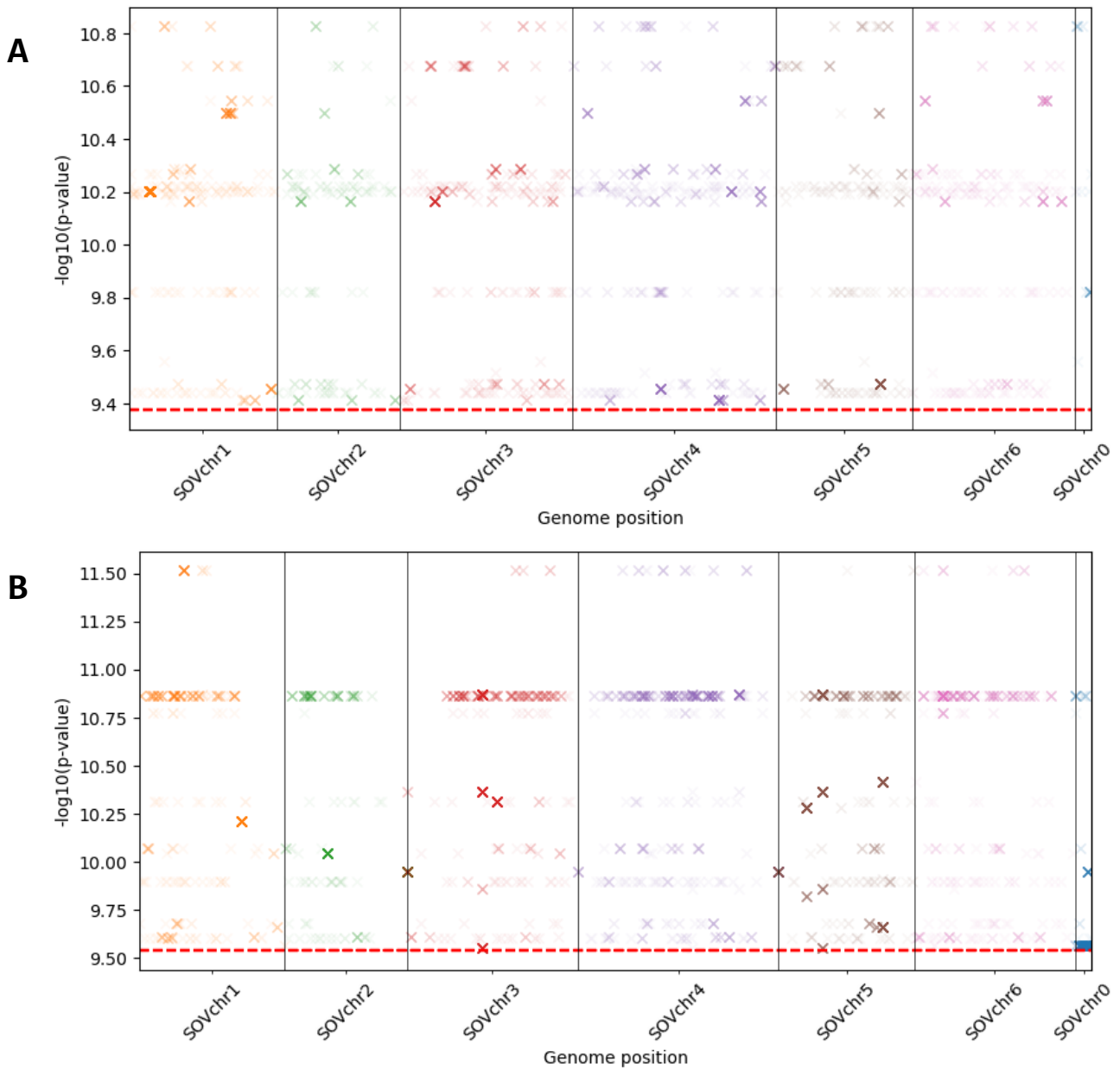


Figure 24: K-mer-based Manhattan plot of GWAS results for (A) leaf shape and (B) leaf apex shape. Displayed are the lowest p -values of significant k -mers contained in extracted sequencing reads and their corresponding genomic mapping positions, including unanchored scaffolds (collected in “SOVchro”). The dashed red line indicates the genome-wide significance threshold for a 5 % family-wise error rate. The higher the opacity of a data point, the higher the number of supporting mapped reads.

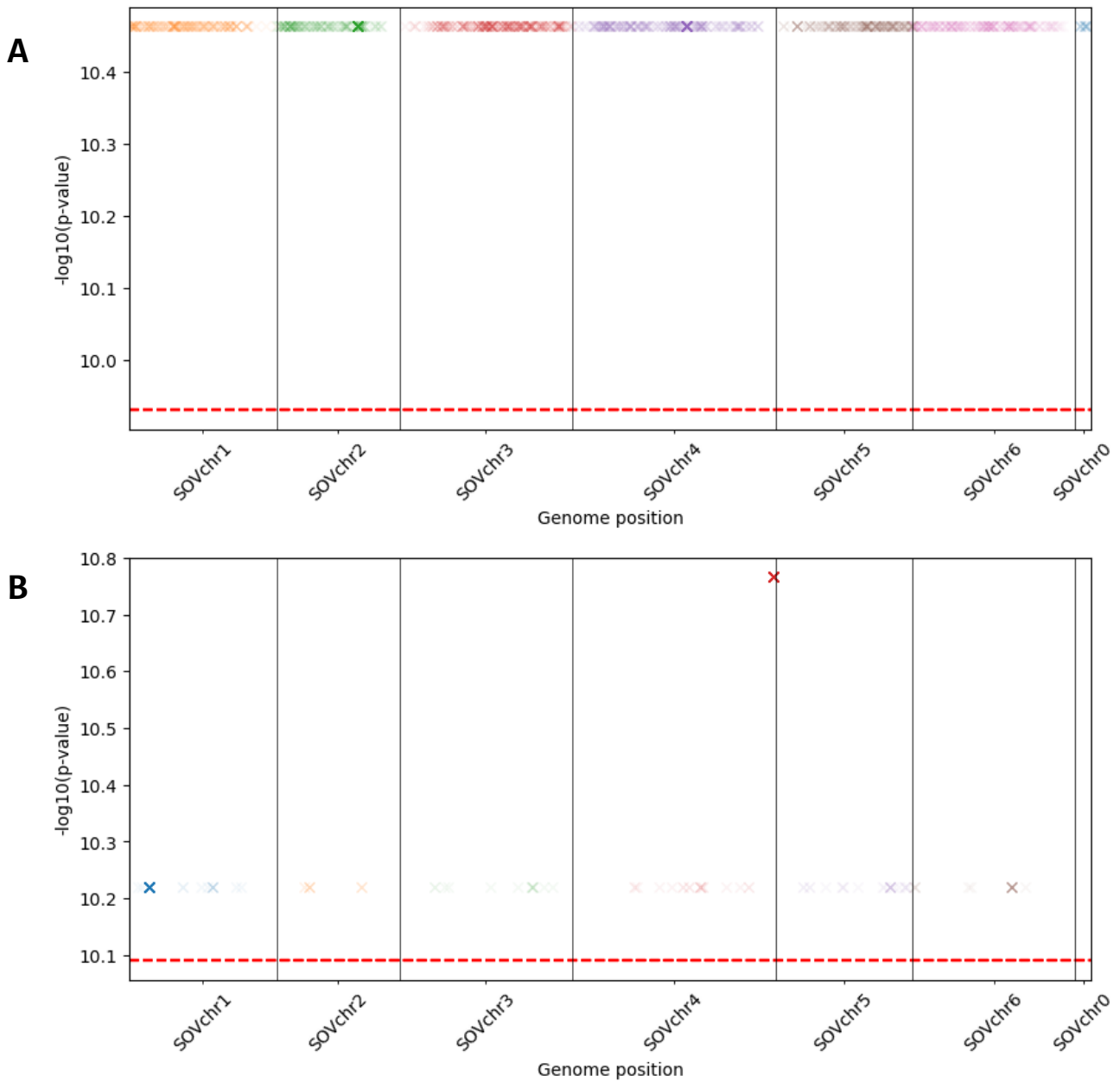


Figure 25: K-mer-based Manhattan plot of GWAS results for (A) flowering days and (B) bolting days. Displayed are the lowest p -values of significant k -mers contained in extracted sequencing reads and their corresponding genomic mapping positions, including unanchored scaffolds (collected in “SOVchro”). The dashed red line indicates the genome-wide significance threshold for a 5 % family-wise error rate. The higher the opacity of a data point, the higher the number of supporting mapped reads.

Appendix E

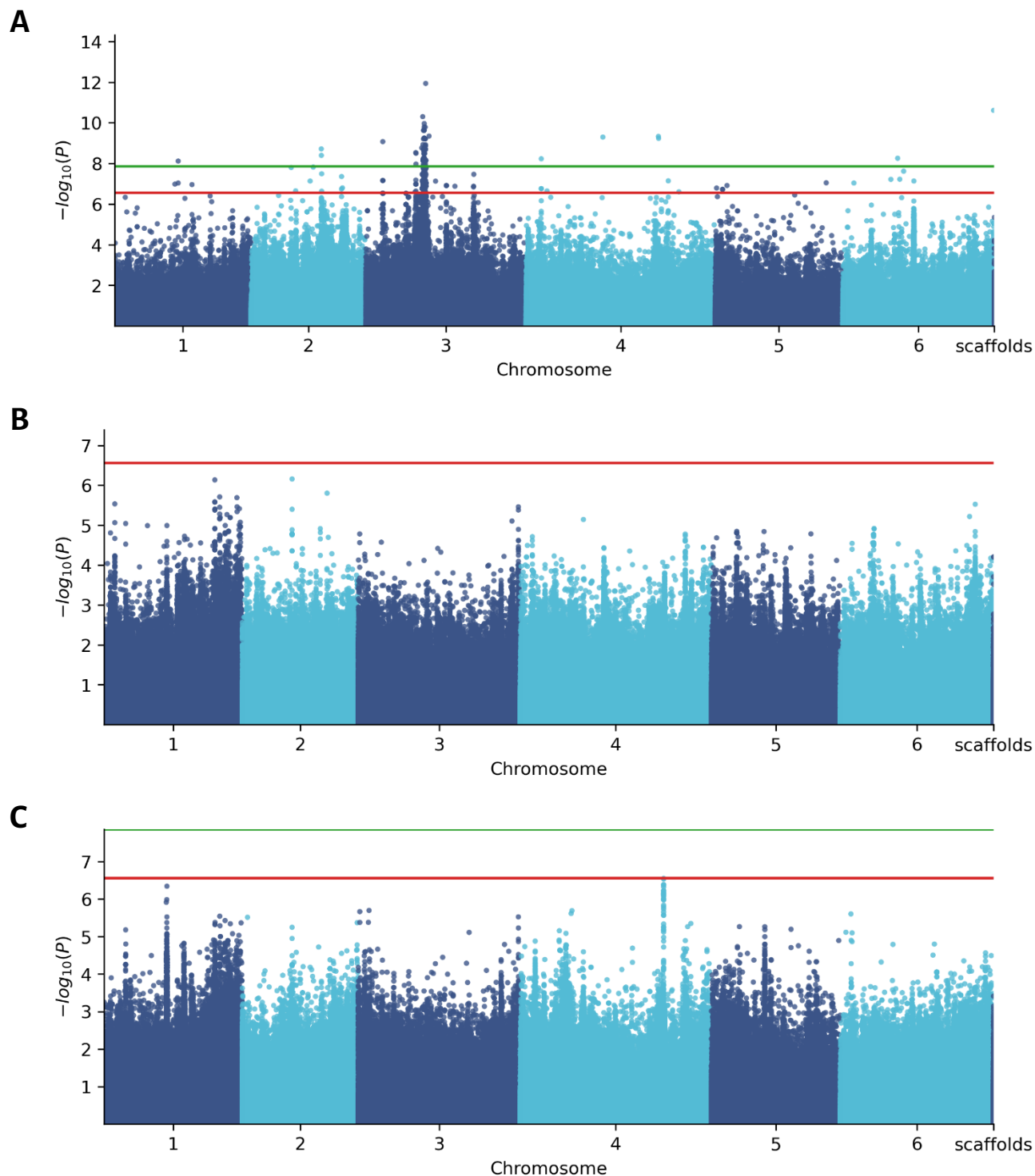


Figure 26: Manhattan plot of GWAS results for (A) plant type, (B) plant height, and (C) plant width. Displayed are the negative logarithmic p -values of SNV-trait associations at their genomic positions. The red and green horizontal lines indicate the suggestive and genome-wide significance thresholds (at $\alpha = 1$ and $\alpha = 0.05$, respectively).

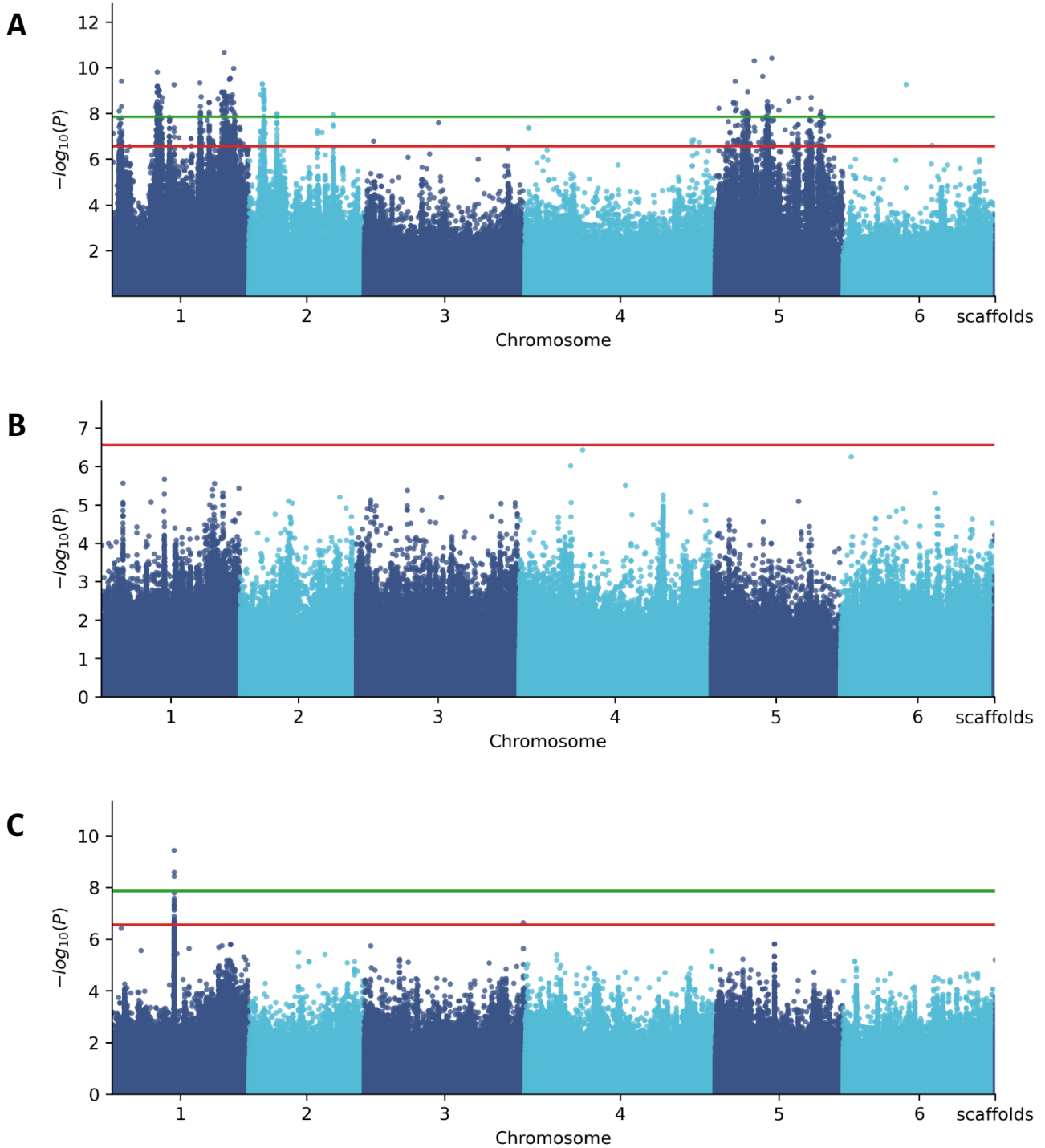


Figure 27: Manhattan plot of GWAS results for (A) leaf number, (B) leaf length, and (C) leaf width. Displayed are the negative logarithmic p -values of SNV -trait associations at their genomic positions. The red and green horizontal lines indicate the suggestive and genome-wide significance thresholds (at $\alpha = 1$ and $\alpha = 0.05$, respectively).

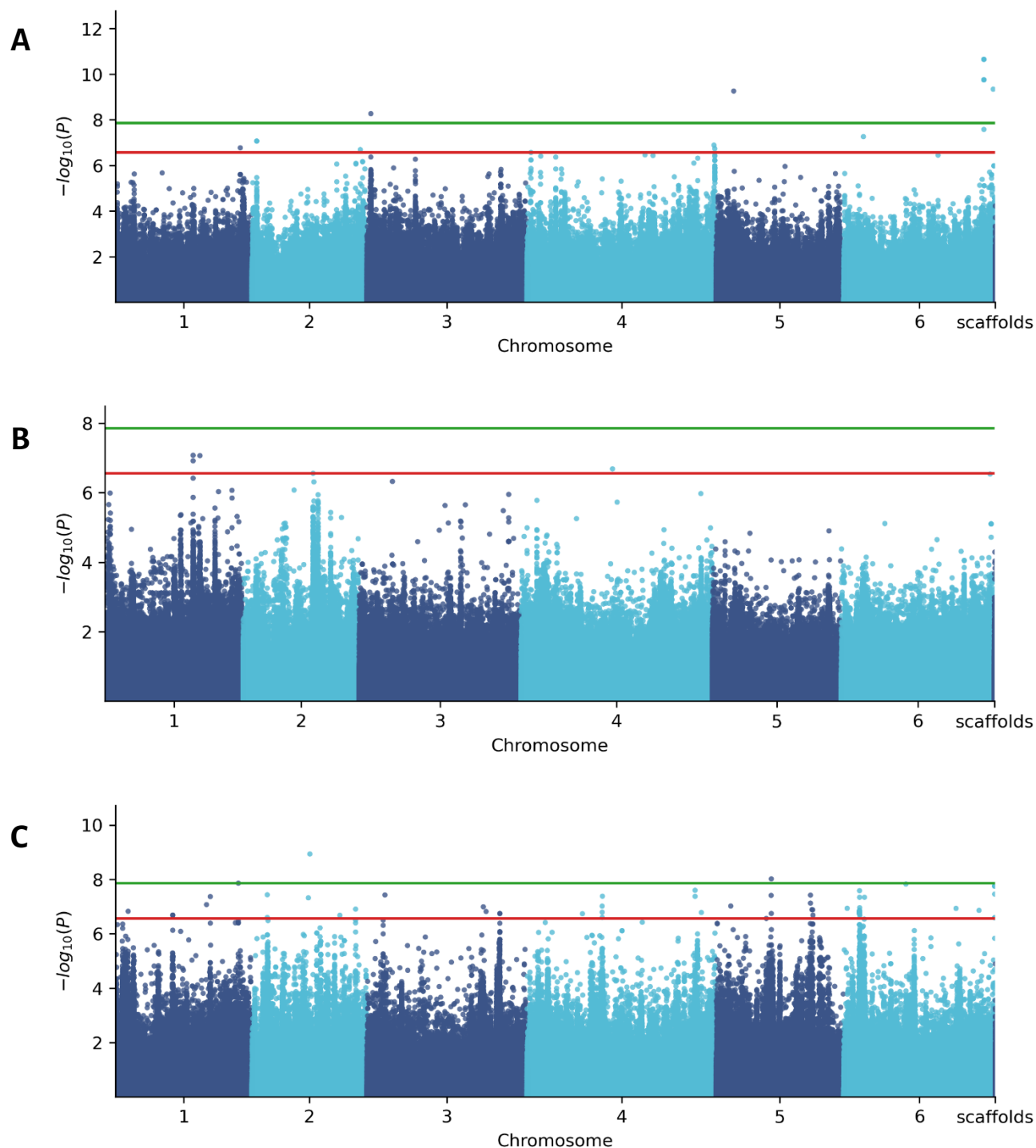


Figure 28: Manhattan plot of GWAS results for (A) leaf surface texture, (B) leaf shape, and (C) leaf apex shape. Displayed are the negative logarithmic p -values of SNV-trait associations at their genomic positions. The red and green horizontal lines indicate the suggestive and genome-wide significance thresholds (at $\alpha = 1$ and $\alpha = 0.05$, respectively).

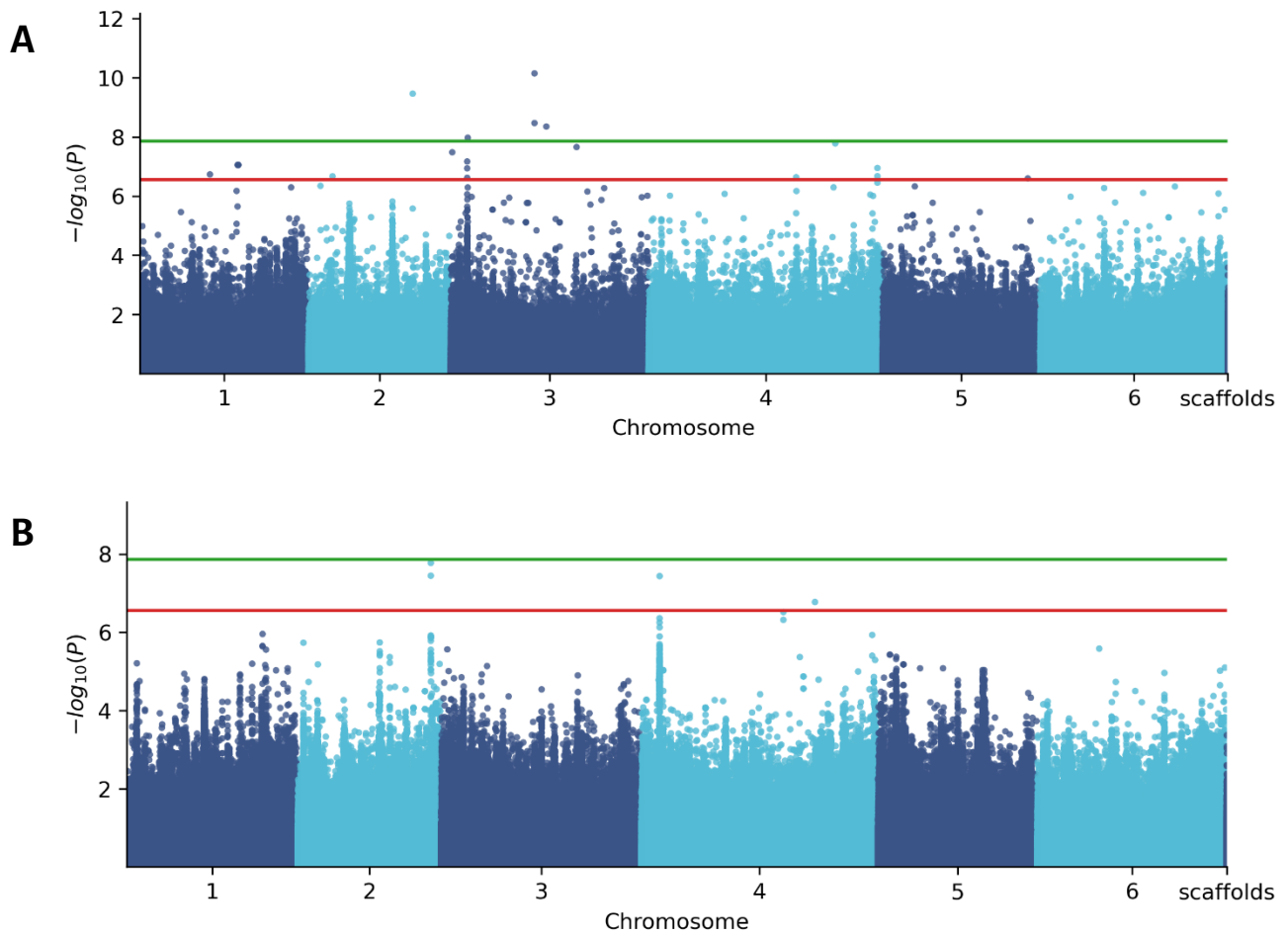


Figure 29: Manhattan plot of GWAS results for (A) leaf base shape and (B) leaf division. Displayed are the negative logarithmic p -values of SNV -trait associations at their genomic positions. The red and green horizontal lines indicate the suggestive and genome-wide significance thresholds (at $\alpha = 1$ and $\alpha = 0.05$, respectively).

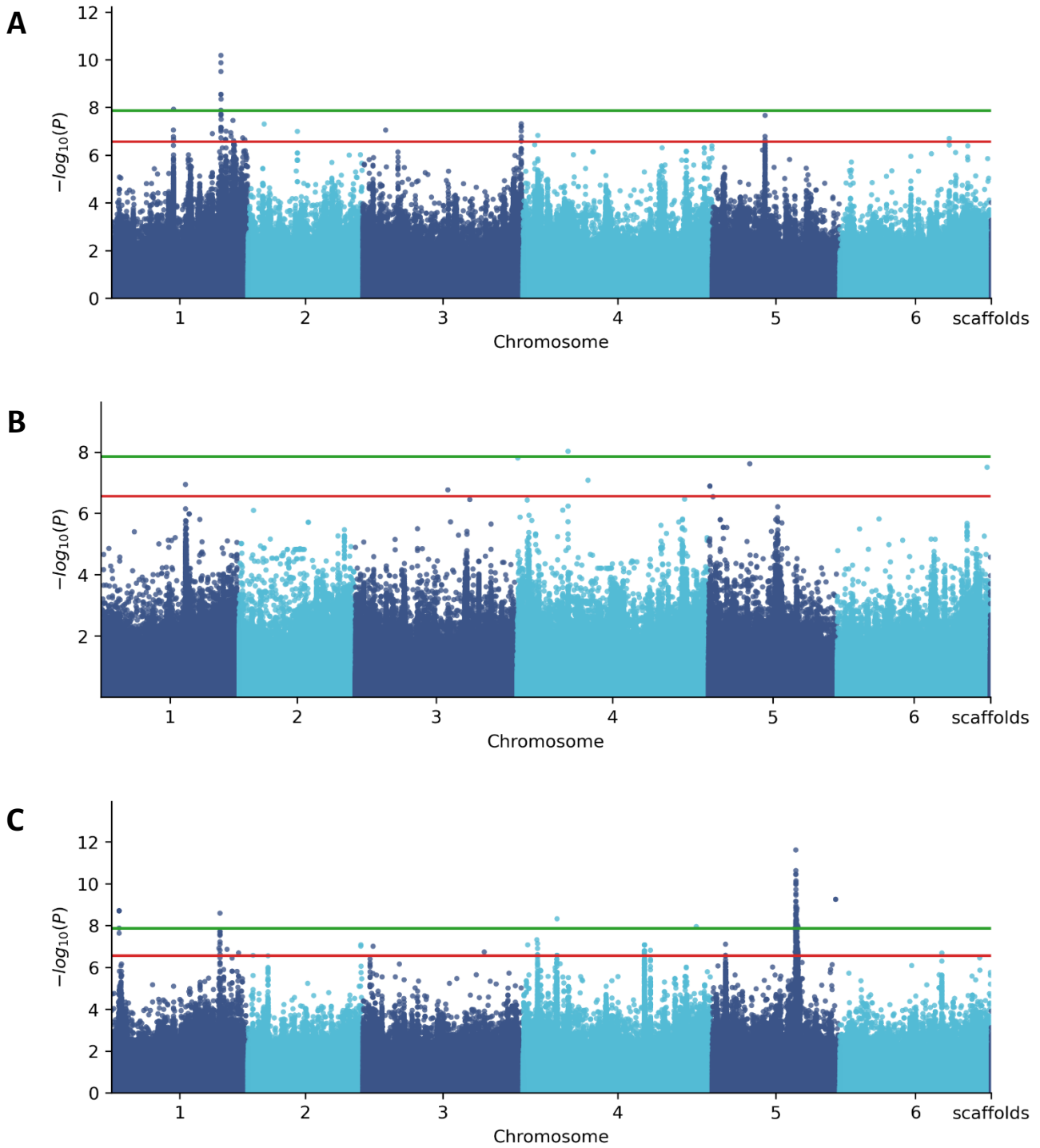


Figure 30: Manhattan plot of GWAS results for (A) petiole length, (B) petiole width, and (C) petiole color. Displayed are the negative logarithmic p -values of SNV-trait associations at their genomic positions. The red and green horizontal lines indicate the suggestive and genome-wide significance thresholds (at $\alpha = 1$ and $\alpha = 0.05$, respectively).

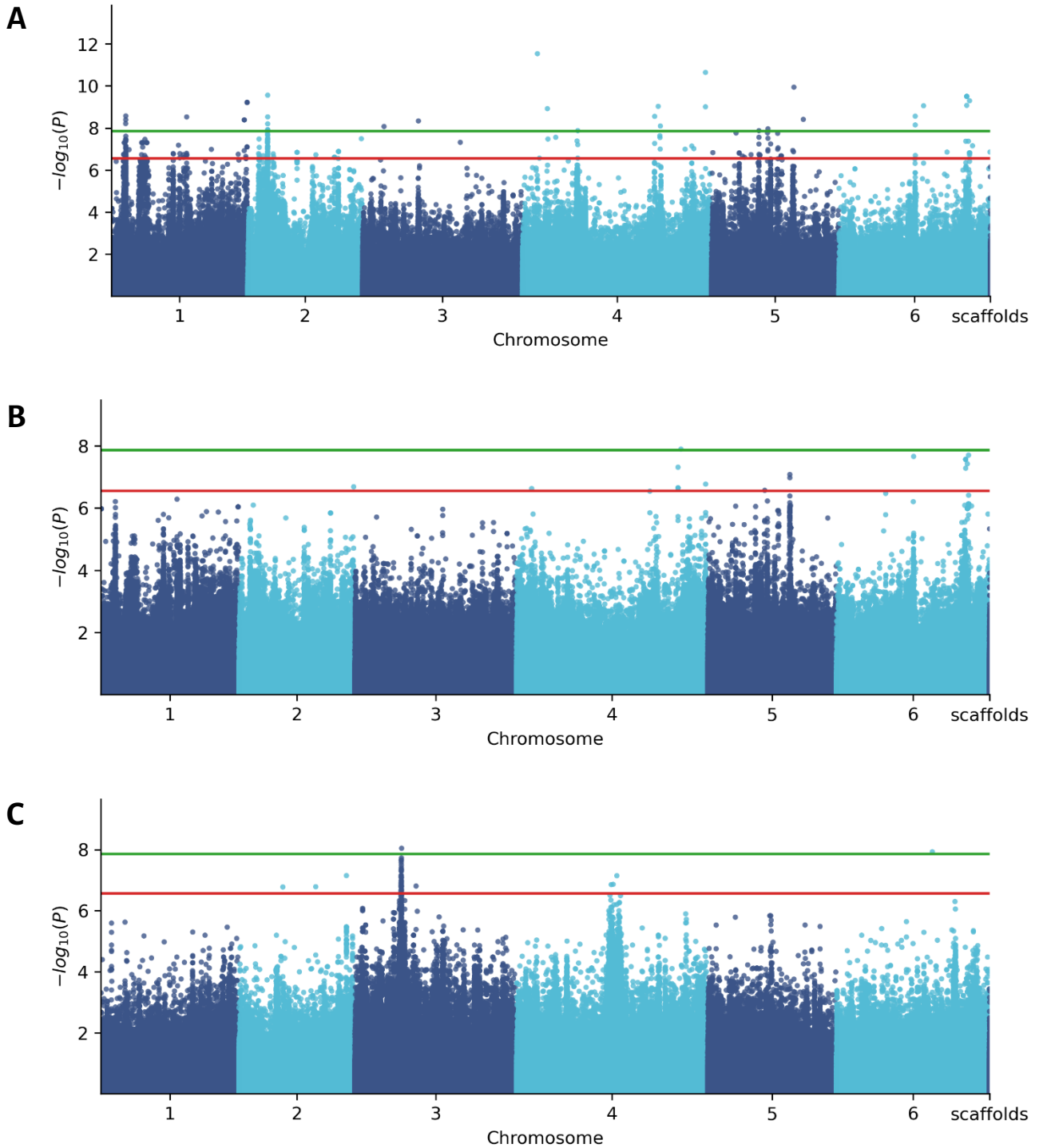


Figure 31: Manhattan plot of GWAS results for (A) bolting days, (B) flowering days, and (C) sex type. Displayed are the negative logarithmic p -values of SNV-trait associations at their genomic positions. The red and green horizontal lines indicate the suggestive and genome-wide significance thresholds (at $\alpha = 1$ and $\alpha = 0.05$, respectively).

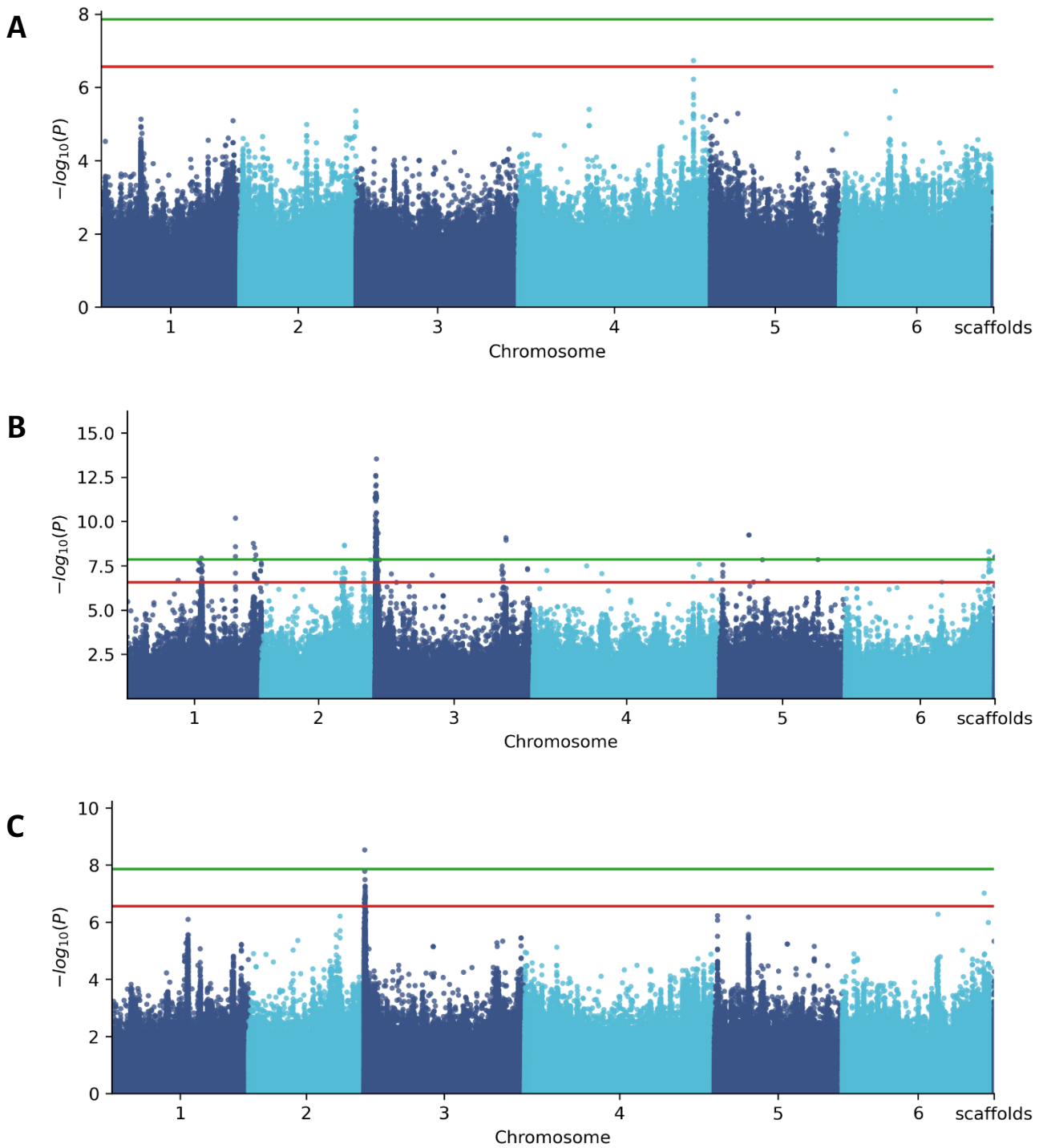


Figure 32: Manhattan plot of GWAS results for (A) oxalate content, (B) downy mildew resistance, and (C) downy mildew incidence. Displayed are the negative logarithmic p -values of SNV -trait associations at their genomic positions. The red and green horizontal lines indicate the suggestive and genome-wide significance thresholds (at $\alpha = 1$ and $\alpha = 0.05$, respectively).